Bitcoin's Required Return

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Abstract

What return should Bitcoin earn? Its history says 32% a year, the CAPM 11–16%, its holders 14–22%. We build the required return from the risks a holder bears: Bitcoin's market risk, its volatility given holders' concentration in it, and its risk of death, from a census of 520 cryptoasset deaths. Evaluated through time and compared in matching units, this required return is an unbiased account of Bitcoin's realized drift since 2012, with a wide band. Today it is 19% a year (90% range 16–22%), about 10% compounded, well below what history extrapolates, and it should keep falling as the holder base broadens.

Keywords: Bitcoin; cryptoassets; required return; expected return; survival analysis; limited participation; marginal holder; portfolio construction.

1. INTRODUCTION

Suppose you must write down one number: the return Bitcoin should be expected to earn, per year, over the next five years. Three candidate answers are on the table, and they disagree by a factor of three.

The first comes from Bitcoin's own history. A simple rule, the exponent of Bitcoin's power-law price trend, about 5.6 (Burger 2019; Papadogiannis 2026c), divided by Bitcoin's age in years, turned out to be the only candidate that forecast the returns that actually followed without bias over 2012–2026 (Papadogiannis 2026c). Applied today it says about 32% a year.[1] The second comes from standard asset pricing. The CAPM asks what Bitcoin's risk would cost a diversified investor: the ten-year Treasury yield of 5%, plus Bitcoin's beta to the stock market times an equity premium of 4% (6% as a high case). Bitcoin's beta to the S&P 500, estimated on month-end total returns, is 1.75 over the last four years, 1.5 since 2016, and 1.8 since 2012 (Appendix A), which gives 11–16% across those windows and the two premia. (The companion paper reported lower betas for the recent windows; Appendix A traces the difference to the alignment of the monthly series and uses the estimates reported here.) The third comes from asking people. In a large US household survey, those who hold crypto said they expected 22% over the next year in 2021 and 13.8%

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[1] Units, once and for all: the age rule's instantaneous (arithmetic mean) drift at age A is $\beta/A \approx 32\%$ at $A = 17.4$; its five-year-average form, used in the tables, is $(1+5/A)^{\beta/5} - 1 \approx 33\%$. The compound (geometric) counterpart is lower by roughly half the variance, $\sigma^2/2$, and so depends on the volatility assumed: a drag of about 4.5 points at 30% volatility and 10 points at 45%, which is why the compound figure is a range, 22–28%, and the arithmetic figure a single number. Section 7 says which form each test uses.

in 2025 (Candia, Coibion, Gorodnichenko, and Weber 2026); 14–22% is the spread between the two dates. Neither range is a confidence interval; each is one quantity read in several ways.

These are not three estimates of one quantity; the disagreement begins with using one word for three things. The first is an *expected return*: a property of Bitcoin's price process, checkable only against what it later earns. The second is a *normative required return*: what an investor holding the CAPM's diversified portfolio ought to demand for Bitcoin's risk, whether or not any such investor holds it. The third is the *required return of the people who actually hold Bitcoin*, revealed by their portfolios and stated expectations; it is the one the price must satisfy in equilibrium, because they are the ones who must be willing to keep holding. The companion paper that documented the first number (Papadogiannis 2026c) closed by posing exactly this puzzle: is the gap between what Bitcoin delivered and what any model says it should deliver *compensation for risks that never materialized*, or *the mark of an asset the market underpriced*? Realized returns alone cannot say, because Bitcoin is one asset with one history, studied precisely because it survived.

This paper answers by measuring the two things the time series cannot identify. The logic is simple. Bitcoin pays no dividend or coupon, so it is worth holding only if its price is expected to rise by enough to pay its holders for the risks they take. Three risks matter, and each has a price: the day-to-day volatility of the asset, which costs more the larger the share of her wealth a holder keeps in it; the systematic risk it shares with the stock market, which has grown as institutions have entered (the correlation with equities was 0.1–0.25 before 2020 and is about 0.5 now; with Bitcoin several times as volatile as the market that is a beta near 1.75, worth about 7 points a year); and the risk that the asset dies outright, through a ban, a protocol failure, or a collapse of the kind that ended Terra and FTX. Add the three to the risk-free rate and you have the return the holder requires. The first two are measurable from Bitcoin's own prices and from surveys of who holds it. The third cannot be measured from Bitcoin, which has not died; it can only be measured from the deaths of other cryptoassets, as a life insurer prices one person from mortality tables of many. That is why a paper about Bitcoin's return spends a third of its length on the deaths of other coins.

First, we measure the risks that never materialized, by looking at the assets for which they did. We build a panel of 1,815 cryptoassets from 2010 to 2026, including 520 that died, and estimate the asset class's mortality curves as a demographer would. (Earlier descriptive studies of "coin death" exist, notably Feder, Gandal, Hamrick, and Moore (2018) and Grobys and Sapkota (2020); new here is a curated, pricing-oriented census: death types, dates, recovery fractions, size- and elite-conditional hazards, and the Lindy test.) Two results matter. The popular "Lindy" intuition, that the longer an asset has survived the safer it is, with hazard falling like one over age, is rejected: the death rate of cryptoassets *rises* with age. And the hazard that matters for Bitcoin, the death rate of assets that reached the top twenty, is small: 3.5% a year by any cause, and 0.30% a year for the sudden, value-destroying collapses (Terra's LUNA and UST and FTX's FTT, all in 2022). Priced at near-total loss, survival risk is worth a few points of expected return; it cannot explain a gap of roughly 27 points (the age rule's 32% less the $\sim$5% risk-free rate).

Second, we derive the return Bitcoin's actual owners must require, from the risks they bear and their own portfolios. The investor whose willingness to hold sets Bitcoin's price is not the diversified investor of the CAPM. The average crypto-owning household held 19% of its financial assets in crypto in 2021; option prices from the same era imply the marginal trader was closer to all-in. For such an investor, elementary portfolio mathematics ties the return Bitcoin must promise to observable quantities. We call the resulting formula the marginal-holder condition (Section 2, equation 1): a market-risk term, a concen-

tration term equal to risk aversion times variance times portfolio weight, and a mortality term. The route is Merton's (1987) limited-participation pricing with every input measured, not a factor regression and not a consumption-based model, neither of which can see the investor who actually clears this market. The required return was enormous when Bitcoin's volatility was 120% and its holders were concentrated, and it has fallen, term by measurable term, to about 19% today (90% range 16–22%).

With both measurements in hand, the required return can be evaluated at every date since 2012 and set against what Bitcoin then delivered, in the same units: the required return is an arithmetic drift, so the comparison is with the realized drift, not the compound return, which at Bitcoin's volatility lies tens of points lower. On that comparison the required return is an unbiased account of history: over the five-year windows starting 2012–2021 the realized drift exceeded the midpoint of the required band by 5 points a year on average, against levels of 30–150%, and since the 2021 survey Bitcoin has delivered a drift of 27% against a requirement of 23–47%. The band is wide, because the early marginal holder is not identified, and it does not track the timing of booms and busts, which the smooth age rule fits better in sample. But its level is right on average, which is the test a cost of capital has to pass, and it is built from risks rather than fitted to returns. The age rule, on this reading, is a reduced form of a required return that was itself falling; it now sits above the band.

What this means in practice. In plain terms, the findings are these. Bitcoin's price has to rise by about 19% a year on average, about 10% compounded, to keep its current holders willing to hold it; that is the return its risk justifies today, and it has fallen from well over 100% a year when Bitcoin was volatile and held by a few concentrated enthusiasts. The high returns of Bitcoin's past were, on average, what its risks required at the time, not a free lunch and not a law of nature. For an allocator, three things follow. Extrapolating Bitcoin's history (the 32%) overstates its prospective return by about two-fifths. Bitcoin is no longer uncorrelated with equities: about 7 of the 19 points are compensation for stock-market risk the allocator already holds, and a diversified investor's hurdle is 12–15%. And the position size the evidence supports for a moderately risk-averse investor who accepts the 19% is on the order of 10% of financial wealth, far below what Bitcoin's history would suggest; for smaller cryptoassets the same arithmetic says their holders have not been paid for the risks they carry.

The paper's contributions, in order: (i) mortality curves for cryptoassets built for pricing, with the Lindy hypothesis tested and rejected and the top-asset hazard measured; (ii) a required-return estimate for Bitcoin built on Merton's (1987) limited-participation condition, with the holder weight from household surveys, a rolling market beta, and the mortality term from the census; (iii) a comparison of that required return with Bitcoin's realized drift in matching units, which it passes on average; and (iv) a dated, falsifiable forecast: about 19% a year in drift, with a 90% range of 16–22%, about 10% compounded, for September 2026–2031, against the age rule's 32%. As checks, the same condition is applied to Ether, to smaller cryptoassets, and (Appendix E) to gold, where it reproduces a known result.

Section 2 sets out the marginal-holder condition, in words first. Section 3 describes the data. Section 4 presents the graveyard. Section 5 assembles what holders require today. Section 6 evaluates the required return through time against realized drift. Section 7 states the forecast and what would falsify it; Section 8 applies the same pricing to Ether and to smaller cryptoassets; Section 9 translates the results for an allocator and compares Bitcoin with equities; Section 10 lists limitations; Section 11 concludes. The appendices take the three terms of the equation in turn (A market risk, B concentration, C mortality), then assembly and

validation (D) and extensions (E).

2. TWO RETURNS, AND THE PERSON WHO SETS THE PRICE

Everything in this paper runs through one equation, so we build it slowly and then display it.

Expected versus required. An asset's *expected return* is a forecast: what it will earn on average. Its *required return* is a hurdle: what an investor needs it to earn to be willing to hold it, given its risk. The first is positive, a statement about the price process that only realized returns can check, and they check it badly: Elton (1999) shows that realized returns are so noisy a proxy that decades of data can fail to distinguish competing values. The second comes in two forms that are often run together. The *market's* required return is what the marginal holder actually demands, a positive fact about people rather than prices, revealed by portfolio choices (Giglio, Maggiori, Stroebel, and Utkus 2021), stated expectations (Greenwood and Shleifer 2014), and option prices (Martin 2017). The *normative* required return is what an investor with stated preferences ought to demand, derived from a model; the CAPM number is of this kind (Sharpe 1964; Lintner 1965): a hurdle, not a description of anyone. In equilibrium the expected return equals the market's required return: the price adjusts until the forecast equals what the marginal holder demands. That equality is the whole content of "the market is in equilibrium"; it is why Cochrane (2011) can treat expected returns and discount rates as one object, and testing it is what empirical asset pricing does (Fama and French 2004). Forecasters work on the left side of the equality and financial economists on the right; the subject lives in the gap. For an asset whose holders are still learning what it is, the gap can stay open for years, because learning makes realized returns predictable in ways no required return explains (Lewellen and Shanken 2002; Pástor and Veronesi 2009a); the separation of the two returns for valuation is developed in Papadogiannis (2026a, 2026b).

In these terms the age rule estimates the expected return, the CAPM figures are normative required returns, and holders' surveyed expectations are evidence on the market's required return. The surveyed expectations are a number without a model behind it; this paper builds the model, deriving the market's required return for Bitcoin from the risks its holders bear, with every input measured, and checking the result against what holders say and what options imply. It uses the normative form only for the diversified allocator of Section 9, and then tests the result where a cost of capital should be tested, against what the asset delivered (Section 6).

Whose hurdle? Here is the step most discussions skip. An asset has as many required returns as it has potential investors, but only one of them is reflected in the price: the hurdle of the *marginal holder*, the investor who is just indifferent between holding a little more and a little less. If a pension fund would be happy to hold Bitcoin at an expected 8%, but pension funds are not allowed to hold it, their 8% is irrelevant to the price. The price is set by whoever actually clears the market.

What risk aversion means. Everything rests on one assumption about people: an extra euro is worth less to you when you are richer. Risk aversion, γ, is the rate at which that happens: if your wealth rises by 1%, the value you place on one more euro falls by γ%. With $\gamma = 0$ a euro is a euro whatever your wealth, so you would take any fair bet; with $\gamma = 2$ each extra euro is worth 2% less for every 1% richer you become; with $\gamma = 4$, 4% less. The larger γ, the more a loss hurts relative to how much an equal gain helps. This is the Arrow–Pratt coefficient of relative risk aversion, and "constant" relative risk aversion means it does not change with wealth. For gambles the consequence is direct: a fair gamble leaves such a person worse off,

and the payment she needs to accept it is about $\gamma/2$ times the gamble's variance, as a share of wealth. So $\gamma = 2$ means paying roughly 1% of wealth to avoid a coin flip over $\pm 10\%$ of wealth; $\gamma = 1$ (log utility) pays 0.5%, $\gamma = 4$ pays 2%. The range 1 to 4 is not a convention but where estimates keep landing: about 2 from US household asset holdings (Friend and Blume 1975), below roughly 2 from labor-supply behavior (Chetty 2006), about 1.7–2 from Italian household panels, which also support the constant-relative form (Chiappori and Paiella 2011), and small single digits in the macro calibrations that followed Mehra and Prescott (1985); 2–3 is the workhorse range.

The marginal holder's arithmetic. Suppose this investor holds a fraction w of her wealth in an asset with annual return volatility σ whose return is for the moment uncorrelated with the rest of her portfolio (Bitcoin's correlation with equities stayed below 0.25 until 2020, as in Liu and Tsyvinski 2021, and has since risen to about 0.5; the market term added below carries that part). Should she hold a little more? The benefit is the asset's expected return above the risk-free rate, the *excess return*, since the euro could otherwise sit in a Treasury bill; that risk-free rate is the baseline for every return in this paper. The cost is the extra risk added to her portfolio, which, because the rest of the portfolio is uncorrelated with the asset, is proportional to how much she already holds, w, times its variance, σ^2. Her price for bearing risk is γ, so the cost in return units is $\gamma\sigma^2 w$. (The $\gamma/2$ of the gamble example has not gone missing: it prices the whole position, whose variance as a share of wealth is $\sigma^2 w^2$, at a total cost of $\gamma\sigma^2 w^2/2$; the decision here is about one more euro, so the relevant cost is the marginal one, the derivative of that total with respect to w, which is $\gamma\sigma^2 w$.) She stops adding when benefit equals cost: expected excess return $= \gamma\sigma^2 w$. The three multipliers are symmetric: halving her weight lowers the return she requires exactly as much as halving her risk aversion or the variance would, which is why a dilution of the holder base, as in the ETF era, lowers the required return just as a fall in volatility does. With the inputs measured later, Bitcoin's current volatility $\sigma = 0.42$ (Section 5), so $\sigma^2 \approx 0.18$, and the surveyed share of crypto in holders' financial assets, $w = 0.19$ in 2021 (Section 5), a holder of middling risk aversion $\gamma = 2$ requires $0.18 \times 0.19 \times 2 \approx 6.7$ points a year above the risk-free rate for the volatility alone; with $\gamma = 4$, twice that; a holder with only 5% of her wealth in Bitcoin, 1.8 points. Read one way this is a sizing rule: given the return you expect and your γ, hold $w =$ excess return$/(\gamma\sigma^2)$. Read the other way, which is what this paper does, an observed weight and an observed volatility reveal the excess return the holder must have been requiring. Adding the two risks this simplest version leaves out, market risk and sudden death, gives the central equation of the paper, which we call the **marginal-holder condition**. It is Merton's (1969) portfolio first-order condition, used as a pricing statement for an asset that only some investors hold as in Merton (1987), with the compensation for a jump to ruin from Merton (1976):

$$\boxed{\mu - r_f \;=\; \underbrace{\beta\,\mathrm{ERP}}_{\text{market risk}} \;+\; \underbrace{\gamma\sigma^2 w}_{\text{concentration}} \;+\; \underbrace{\lambda\,(1-\delta)}_{\text{mortality}}} \tag{1}$$

Read the terms left to right. The left side is the asset's expected return over the risk-free rate. On the right: a CAPM-style term for the market risk the asset carries (beta times the equity risk premium); the **concentration term**, γ times variance times the holder's own portfolio weight, which prices the risk she cannot diversify because she chooses, or is forced, to hold a lot of one thing; and a **mortality term**, the probability per year that the asset dies, λ, times the fraction of each euro lost when it does, $(1-\delta)$. (Every term is stated per euro held in the asset; the holder's portfolio as a whole loses $(1-\delta)$ times her weight w.)

The mortality term, carefully. The two symbols answer different questions. λ is the probability per year that the death event occurs. δ is the *recovery fraction*, the share of her money a holder keeps when it does, so $(1-\delta)$ is the share she loses. They must be multiplied because dying and losing everything are not the same thing. When TerraUSD collapsed, holders kept about two cents on the dollar ($\delta \approx 0.02$). When BUSD was wound down under regulatory order, a death by any institutional definition, holders redeemed at par: $\delta \approx 1$, and nobody lost a cent. When a token merely migrates tickers, $\delta = 1$ exactly, which is why Appendix C does not count migrations as deaths. The product $\lambda(1-\delta)$ is the *expected loss rate from mortality*, and it belongs in the drift because over a short interval dt the holder earns the surviving drift $\mu\,dt$ with probability $1-\lambda\,dt$ and loses $(1-\delta)$ with probability $\lambda\,dt$, so her expected return counting both outcomes is $\mu\,dt-\lambda(1-\delta)\,dt$; setting that equal to the required $r\,dt$ gives $\mu = r+\lambda(1-\delta)$. The price must climb, while the asset lives, by exactly the expected death loss, an actuarially fair insurance premium paid in drift, so that averaged over survival and death the holder earns what she requires. A 3% annual death probability with 95% lost on death costs $0.03 \times 0.95 \approx 2.9$ points a year. This is also the precise sense in which a backtest run on a surviving asset overstates the expected return: it measures μ, the drift on the paths where death did not happen, while the unconditional expected return is $\mu-\lambda(1-\delta)$; the crashes that would have pulled the average down are missing from the sample by construction (Brown, Goetzmann, and Ross 1995).

What equation (1) is, and is not. Nothing in equation (1) is new, and that is deliberate. The concentration term is the first-order condition of textbook portfolio choice (Merton 1969), and its use as a pricing statement under limited participation is Merton (1987): when only a fraction of investors hold an asset, those who do must be paid for the idiosyncratic variance they absorb, in proportion to how much each absorbs. The mortality term is the standard jump compensation (Merton 1976) read through the survival conditioning of Brown, Goetzmann, and Ross (1995), and a term of the same form, hazard times loss, appears wherever a portfolio problem includes a jump to ruin, from event-risk allocation (Liu, Longstaff, and Pan 2003; Das and Uppal 2004) to the rare-disaster models that followed Barro (2006). The beta term is the CAPM. What is ours is the assembly and, above all, the mapping: each term matched to something observable for this asset class, the holder's weight to a survey, the death hazard to a census of the asset's relatives, the whole to stated expectations and option prices. To our knowledge that mapping has not been written down for cryptoassets; the contribution is not the equation but its inputs. The mechanism the concentration term describes has a name and an empirical record. Merton (1987) drew the conclusion himself: assets held by few carry a higher cost of capital, and it falls as recognition broadens, the *investor recognition hypothesis*. Levy (1978) had derived the pricing of idiosyncratic variance by investors who hold few securities a decade earlier, and Errunza and Losq (1985) the same form for securities that foreign investors cannot hold. The hypothesis has been tested for individual stocks, where broadening the shareholder base raises the price and lowers subsequent returns (Kadlec and McConnell 1994; Foerster and Karolyi 1999; Amihud, Mendelson, and Uno 1999; Lehavy and Sloan 2008; Bodnaruk and Östberg 2009), and for national markets opening to foreign capital (Bekaert and Harvey 2000; Chari and Henry 2004). Pástor and Veronesi (2009b) give it a lifecycle: a new technology's risk is idiosyncratic while it is small and held by few, and turns systematic as it is adopted. Ibbotson, Idzorek, Kaplan, and Xiong (2018) generalize it into a theory of popularity, in which hard-to-diversify assets earn a premium that popularity erodes. For cryptoassets the record is indirect: a broadening investor base is linked to idiosyncratic volatility (Izadyar and Zamani 2022), and institutional

participation, the spot ETFs in particular, raised Bitcoin's correlation with equities (Copestake, Furceri, and Terracciano 2026), as index investment did for commodities (Tang and Xiong 2012). What the record lacks is the size of the mechanism for an asset class: every test above is a sign test on prices or correlations. Measuring the level, with the holder weight from surveys and the death hazard from a census, is what this paper does. Three clarifications. First, equation (1) is an optimality condition that any equilibrium must satisfy, not an equilibrium model: we do not model supply, endogenize w, σ, or λ, or derive the price level. It complements rather than competes with the equilibrium models of Bitcoin's price (Biais, Bisière, Bouvard, Casamatta, and Menkveld 2023; Pagnotta 2022; Cong, Li, and Wang 2021), supplying the discount-rate input they leave to be estimated. Second, the condition is generic by design: the same line prices founder stock, private equity, and emerging-market equities before liberalization. Bitcoin enters through the values of the inputs, not the functional form, and through one data advantage no other asset class offers: a thousand-asset graveyard from which λ can be measured rather than assumed. The emerging-market integration literature (Bekaert and Harvey 2000; Henry 2000) documents the mechanism our forecast rests on: required returns are high while a market is segmented and compress when it opens to outside capital; the ETF era is Bitcoin's liberalization. Third, using the survey-average holder weight for w is not casual: in an equilibrium of holders with similar risk aversion, every unconstrained holder's condition binds at the same $\gamma\sigma^2 w$, so weights equalize across participants and the average participant's weight identifies the marginal one's. The identification assumes that weights are chosen optimally under beliefs that are on average correct; a weight inflated by over-optimism satisfies the same first-order condition but recovers a belief rather than a fair premium, and only the comparison with realized returns in Section 6 can tell the two apart. Fourth, equation (1) is complete only for the investor it describes: a holder with constant relative risk aversion over wealth, facing diffusive risk plus one kind of jump, whose other wealth is the market. (Strictly, the market term carries a factor $(1-w)$ and the equity premium should be the holder's own price of market variance; at w of 8–19% the first is a correction of a point or less, and Appendix A discusses the second.) It is not a complete list of everything that could be priced. What it leaves out has known signs. Pushing the required return up: a liquidity premium (Amihud and Mendelson 1986), material in Bitcoin's early years and small today; a jump-risk premium above the actuarial loss if deaths are systematic, which Section 9 charges for; ambiguity about the distribution itself, natural for a new asset; hedging demands against state variables such as monetary policy, to which Bitcoin has become sensitive (Copestake, Furceri, and Terracciano 2026). Pushing it down: a convenience or service yield from holding the asset, the analogue of gold's unobserved dividend in Barro and Misra (2016); a lottery preference for skewness (Barberis and Huang 2008); and disagreement under short-sale constraints, which raises prices and lowers expected returns (Miller 1977). The paper's defense is not that these are zero but that two independent closures leave little room for them: whatever the market prices for a fully invested holder, of any kind, is in the risk-neutral variance of options, and that bound (16–30 points of excess return for an all-in holder) sits above the 14 points of excess return the paper's 19% implies for a holder at 8–19%, as it should; and the required return so built is unbiased for realized drift over 2012–2021 (Section 6), leaving no large average residual to attribute. Stated expectations are a weaker witness: the 2021 survey answer is reproduced at plausible risk aversion, the 2025 answer is below the model (Section 5). Anything still unpriced shows up as a gap between realized and required, and the forecast tests whether it is there. Every empirical section measures one term: $\lambda(1-\delta)$ from the graveyard (Section 4); σ, w, and γ from prices, surveys, and options (Section 5). Section 6 then evaluates

the condition date by date, Section 7 uses it as a forecast, Section 8 applies it to Ether and other coins, and Section 9 inverts it into a sizing rule.

A worked example. In 2021, Bitcoin's volatility was about 80% a year, so $\sigma^2 = 0.64$. Survey data (Section 5) show the average crypto-owning household held $w = 19\%$ of its financial assets in crypto. With $\gamma = 2$ the concentration term is $2 \times 0.64 \times 0.19 \approx 24$ points a year. Add the risk-free rate of the time, 1.5%, a beta term of about 5 points (the 2021 rolling beta of 1.3 on a 4% equity premium), and 0.3 points of mortality (Section 4), and the marginal holder of 2021 required about 31% a year. Holders surveyed that year said they expected 22%, which the same arithmetic delivers at $\gamma \approx 1.3$, inside the ordinary range.

What makes each term fall. Volatility falls as markets deepen: Bitcoin's σ has gone from about 150% (2013) to about 40–50% (2026). The weight w falls as participation broadens: when the only people willing to hold an asset are enthusiasts, each must hold a lot of it; when ETFs let everyone hold a sliver, nobody has to. And λ falls, or at least the market's estimate of it falls, as the asset survives attacks it was feared not to. One term moves the other way: integration with mainstream portfolios raises the correlation with equities, and with it the market term. Each term is measurable, the sum has fallen, and today it is about 19% a year; Figure 6 in Section 6 shows the path of each.

3. DATA

The asset panel comes from the Coin Metrics community dataset (Coin Metrics 2026), the same source the companion backtest paper uses for Bitcoin and Ether prices, observed from July 2010 to May 24, 2026. After removing fiat exchange rates, tokenized equities, wrapped and staked duplicates, stablecoins, and trading-pair artifacts, the core universe is **1,815 cryptoassets**, contributing about 67,700 asset-months of exposure. The dataset carries full daily *trading volume* for essentially all of them, and full daily *prices* for the 139 historically significant assets; volume therefore carries the life-and-death timeline, and prices carry the return calculations on the significant subset.

Birth. An asset is born, for our purposes, on the first day it shows a price or trading volume: the day it enters the investable, tracked universe. This is the clock an investor cares about.

Death. Death is harder, because cryptoassets rarely die cleanly. We use three mechanical definitions and then a curated layer:

1. **Feed death**: the data feed ends more than 180 days before the dataset does. This proxies delisting from tracked markets. 234 cases.

2. **Price death**: the price falls below 1% of its all-time high and never recovers (measurable only on the price-complete subset). 33 cases.

3. **Volume death**: trailing three-month volume falls below 1% of its historical peak *and* below an absolute floor of $250,000 a day, and never recovers. The absolute floor matters: without it, healthy mid-caps whose volume merely normalized after the 2021 mania would be falsely counted as dead. 438 cases, overlapping the other two definitions.

4. **Curation.** Two corrections are applied by hand, with the full audit trail in Appendix C. *Ticker migrations are not deaths*: when MATIC became POL, EOS became Vaulta, or Golem's GNT became GLM, holders received successor tokens at full value, so these events (seventeen in all) are removed.

And *known catastrophes are dated correctly*: the free data tier hides the price history of several major assets, so the collapses of LUNA and UST are entered at May 12, 2022 and FTT at November 10, 2022 from the public record, rather than at whatever date a volume rule would eventually notice them.

The result: **520 deaths among 1,815 assets (29%)**, of which 8 are classified as sudden, value-destroying *jump-collapses* and the rest as gradual abandonment: projects that faded, lost their listings, and stopped trading.

The other inputs are: Bitcoin's monthly price, market capitalization, and count of addresses with a balance (Coin Metrics); US household crypto-ownership rates from the Federal Reserve's payment and household surveys and Pew (0.4% of adults in 2014, 1.2% in 2018, 9–12% in 2021, roughly 11–14% since); holders' portfolio shares and expectations from Candia, Coibion, Gorodnichenko, and Weber (2026), with Austrian household-finance data (OeNB 2025) as a European cross-check; ten-year Treasury yields; global household wealth from the UBS Global Wealth Report; option-implied figures from Deribit as reported in the literature and the DVOL index; month-end total returns of the S&P 500, represented by the Vanguard S&P 500 ETF (Yahoo Finance); spot-ETF and public-company Bitcoin holdings from issuer disclosures and corporate filings (as compiled by Bitcoin Treasuries 2026); and, for the gold comparison of Appendix E, LBMA gold prices and World Gold Council above-ground stock estimates.

One bias deserves a sentence up front. Coin Metrics adds an asset to its universe only once the asset has become significant, so thousands of micro-cap tokens that died young and obscure never enter the panel. This *understates* the death rate at young ages, which means it biases the test *against* our headline finding that hazard rises with age. The elite estimates, which are what Bitcoin's pricing needs, are unaffected: once an asset is in the top twenty it is fully observed from then on.

4. THE GRAVEYARD: HOW CRYPTOASSETS DIE

A *hazard rate* answers a simple question: of the assets alive at a given age, what fraction die within the next year? If 100 three-year-old coins are alive in January and 12 are dead by December, the hazard at age three is 12% a year. Estimating hazards across ages, with assets that are still alive treated correctly as "censored" (we know they lived *at least* this long), is standard survival analysis; the machinery is the same used for patients, firms, and light bulbs.

The Lindy question. The age rule's most natural economic defense was mortality: perhaps Bitcoin's enormous drift was fair payment for a risk of sudden death that declines as the asset ages. This is the "Lindy effect," under which remaining life expectancy grows with age, so the hazard falls like k divided by age. If that were true with k near 5.6, the age rule would simply be survival compensation, rationally priced, with equation (1)'s mortality term doing all the work. The hypothesis is attractive, parameter-free in shape, and testable; testing it is the point of building the graveyard.

It fails. Table 1 shows the death rate by age bucket under the curated definition.

The hazard *rises* with age through at least age nine. Formally, a Poisson regression of monthly deaths on log age, with calendar-year fixed effects to absorb market-wide waves such as crypto winters and standard errors clustered by asset, gives an age elasticity of **+0.42 (s.e. 0.04)**: each doubling of age *raises* the death rate by roughly a third, holding the calendar year fixed. Lindy requires -1. The estimate has the wrong sign by eleven standard errors, and it is robust: $+0.44$ to $+0.56$ across the three mechanical death definitions,

Table 1: Mortality hazard by age, curated death definition (1,815 assets, 520 deaths, 2010–2026).

Age (years)	Deaths	Exposure (asset-yrs)	Hazard (per yr)
0.1–0.5	6	693	0.9%
0.5–1	37	770	4.8%
1–2	111	1,213	9.2%
2–3	106	905	11.7%
3–4	82	661	12.4%
4–5	78	488	16.0%
5–7	63	502	12.6%
7–9	33	198	16.7%
9–12	4	53	7.5%

Buckets begin at age 0.1 (the first weeks of a listing are definitionally alive) and the sparse 12–17 bucket (7 asset-years, no deaths, Bitcoin the main occupant) is omitted; total exposure across all ages is about 5,640 asset-years.

+0.65 when the suspicious cluster of mid-2025 feed terminations is excluded, and positive within the elite subsample alone. Equivalently, the slope of log survival against log age, which equals k if the hazard really were k/age, gives **between about 0.2 and 0.5** depending on the death definition, **against the 5.6** the age rule would need. A 5.6 would have implied that a three-year-old cryptoasset had roughly a 1-in-20,000 chance of reaching Bitcoin's current age; the measured elite hazard, held constant, implies a 14-year survival probability around 0.6. Cryptoassets age like small firms, with a liability of adolescence and obsolescence, not like Lindy classics (Figure 1).

Where the data end. Three caveats belong next to that result. First, in the oldest populated bucket (ages 9–12) the raw hazard falls back to 7.5%, on 4 deaths and 53 asset-years; the rise is established through roughly age nine, the data's support ends near age twelve, and Bitcoin is 17.4. The Lindy rejection is therefore a statement about the asset class over its observed life span, not a measurement of Bitcoin's own current hazard, which is exactly why the pricing below uses the elite hazard *level* rather than an age extrapolation. Second, elite deaths cluster in calendar time (2022, 2024–25), with contagion episodes and delisting purges, not smoothly in age; the three jump-collapses in particular sit within six months of one another and were causally linked (Terra propagated through Celsius to FTX), so the Poisson confidence interval, which assumes independent events, understates the uncertainty. We treat 0.30% as the point estimate of a partly systematic process, which a diversified allocator may rationally charge for in part; Section 9 does so. Third, the 2024–25 delisting purge supplies 12 of the 36 elite deaths; treating every 2025 death as censoring lowers the all-cause elite hazard from 3.5% to 2.4% a year (CI 1.5–3.5%), so the mortality bound of $\leq$3 points used below is conservative under either reading.

Size is what protects, not age. Sorting assets by peak trading volume (Table 2), the hazard falls monotonically and steeply with size.

The last two lines are the ones Bitcoin's pricing needs. Among the 219 assets that ever reached the top twenty by monthly volume, 36 died over 1,017 asset-years of exposure, an all-cause rate of 3.5% a year, and only three of those deaths were the sudden, catastrophic kind that a jump-risk premium prices: LUNA and UST in May 2022 and FTT in November 2022, each a roughly total loss within days. None has occurred since. Excluding four contested cases, assets flagged dead by the drawdown rule but whose chains still operate (ICP, SUSHI, FLOW, NANO; Appendix C), the all-cause rate falls to 3.15% (CI 2.2–4.4%). The

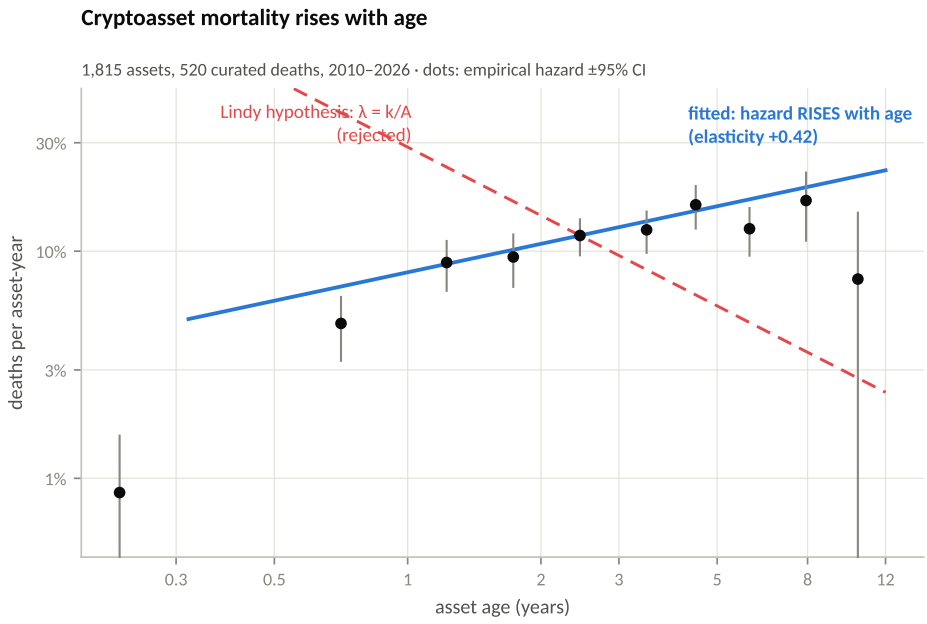


Figure 1: Mortality hazard versus age (log–log). The fitted line uses the period-fixed-effects elasticity of +0.42; the dashed line is the Lindy shape $\lambda = k/A$.

Table 2: Mortality by size and for elite cohorts (curated deaths).

Size group	n	Hazard (per yr)
Smallest quartile	452	17.4%
Q2	451	13.0%
Q3	451	9.8%
Largest quartile	452	3.2%
Ever top-50 by volume	499	5.5% (CI 4.5–6.5%)
Ever top-20 by volume	**219**	**3.5% (CI 2.5–4.9%)**
Ever top-20, jump-collapse only	219	**0.30% (CI 0.06–0.86%)**

remaining elite deaths are slow fades: OMG, NEM, Bitcoin Gold and their like lost their listings over months or years, during which holders could exit at declining prices; losses of that kind live in an asset's volatility and drift, not in its jump hazard. The 520 dead assets are, in the taxonomy of the companion scarcity framework (Papadogiannis 2025), the "physically scarce but no longer desired" quadrant realized at scale: hard-coded supply caps outlived demand, and scarcity without desirability monetizes nothing (Figure 2; survival curves by size in Figure 3; the death timeline, with the 2022 cascade and the 2024–25 delisting purge, in Figure 4).

What this buys the pricing equation. Price the jump channel at near-total loss: $\lambda(1-\delta) \approx 0.3\% \times 0.95 \approx$ **0.3 points a year**. Treat every elite death, even the slow fades, as a jump at total loss and you still cannot get above $\approx$ **3 points**. The mortality term in equation (1) is real (2022 proved it can strike the top ten), but it is a few points, not twenty-seven. One caveat matters: no cross-section can measure Bitcoin's *own* hazard in 2010–2014, when it was the only asset of its kind and plausibly faced existential risk; our bounds apply from the point at which an elite class exists to measure, and Section 6 keeps that early window

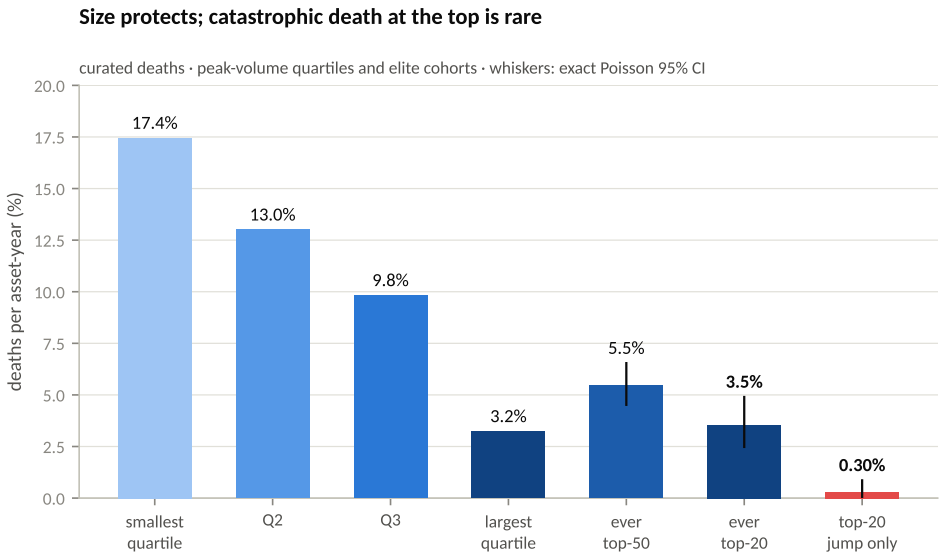


Figure 2: Hazard by peak-volume quartile and for the elite cohorts; whiskers are exact Poisson 95% intervals.

wide.

Risks that have never yet fired for anyone, the clearest being a quantum break of Bitcoin's signature scheme, are λ-events the condition covers by construction but the graveyard cannot price; Appendix C.4 sets out how the option-implied crash intensity referees them and how an investor adds a private view to the mortality term.

5. WHAT BITCOIN'S HOLDERS REQUIRE

The marginal-holder condition, equation (1), needs five inputs. The graveyard supplied the mortality term. This section supplies the holder's portfolio weight and risk aversion, takes volatility from prices and the market beta from Appendix A, assembles the required return for 2026, and sets it against two outside readings, holders' stated expectations and option prices.

Source 1: holders' measured portfolios. The Cleveland Fed study of cryptocurrencies in household finance (Candia, Coibion, Gorodnichenko, and Weber 2026) reports the number the equation needs most: among US households that owned any crypto in 2021Q3, crypto made up **19% of financial assets on average**, with 40% of owners below 5% and about a fifth of owner-households above *half* their portfolio. A European cross-check points the same way: in the Austrian central bank's 2023 household survey, owners mostly held under 20% of financial assets in crypto, rarely above 20–25% (OeNB 2025). So the marginal holder of the early 2020s was not the diversified investor of the CAPM; she held a fifth of her wealth in the asset. For 2025–26 the surveys have not yet published a portfolio share, and the paper uses a range: 19% as the upper bound (the 2021 value, which the market-clearing back-cast of Section 6 also returns for survey-type holders in 2026) and 8% as the lower bound, obtained by adding the spot-ETF and public-company treasury holdings, about 2.5 million BTC by mid-2026, as a second class of holders who are not "crypto owners" in any survey and hold at about 1.5% of their wealth (Appendix B).

Source 2: holders' stated expectations, a partial check. The same surveys asked holders what return

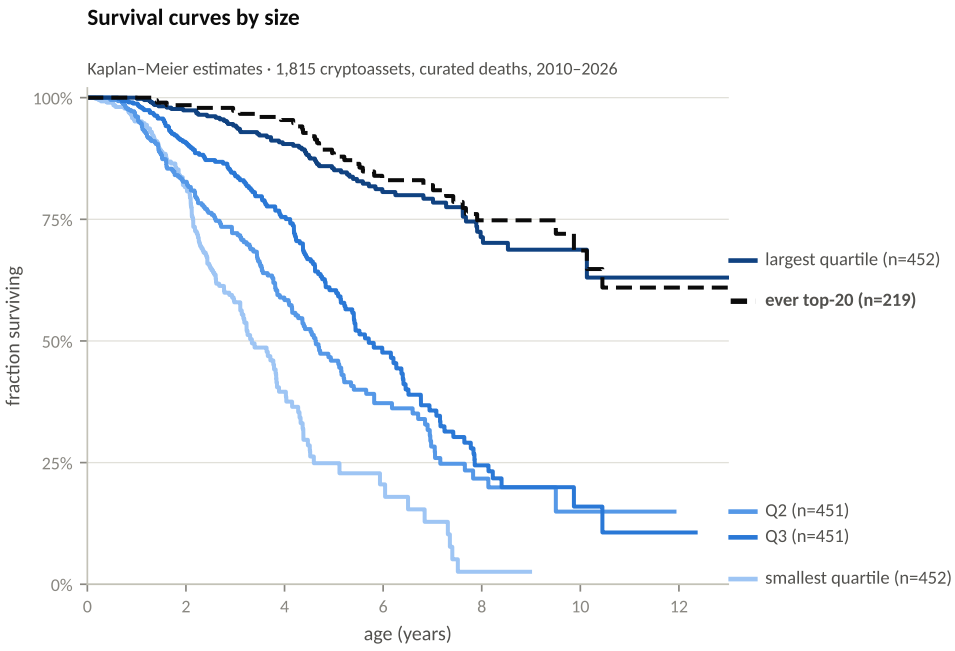


Figure 3: Kaplan–Meier survival curves by peak-volume quartile, with the ever-top-20 cohort (dashed). Half of the smallest quartile is dead by age 3.3; the largest quartile and the elite track each other and remain above 60% survival past age 10.

they expected over the next year: 22% in 2021Q3, 13.8% in 2025Q2. Table 3 sets each answer against equation (1) evaluated with the inputs of its date.

Table 3: Two survey waves, one equation.

	2021Q3	2025Q2
Volatility σ (trailing, at the survey date)	77%	53%
Holder portfolio weight w	0.19 (measured)	0.19 ("similar across waves")
Risk-free rate	1.5%	4.4%
Market term (rolling beta $\times$ 4%)	5.2 pts ($\beta = 1.3$)	7.8 pts ($\beta = 2.0$)
Required return at $\gamma \in [1.7, 2.8]$	26–38%	22–27%
Stated expected return	22%	13.8%
γ that reproduces the stated return	≈ 1.3	≈ 0.2

Mortality term at 0.3 points. Volatility is the paper's trailing estimate averaged over the quarter each wave was fielded; Candia et al. report the 2025 wave as 2025Q2 and the 2025 portfolio share as similar to 2021's. Betas are the rolling four-year estimates of Appendix A at each date.

The table is not a clean confirmation, and we report it as it is. In 2021 holders' stated 22% is somewhat below the model's 26–38% and is reproduced at a risk aversion of about 1.3, inside the ordinary range and close to log utility: beliefs and positions cohere, as Giglio, Maggiori, Stroebel, and Utkus (2021) find for equity investors. In 2025 they do not. Holders said 13.8%, barely more than the risk-free rate plus the market term alone (12.2%), which leaves almost nothing for a fifth of their wealth in an asset with 53% volatility; no plausible risk aversion reproduces the answer. Either holders in 2025 expected less than their own risk required, or they do not perceive the market risk the rolling beta measures, or a twelve-month point forecast

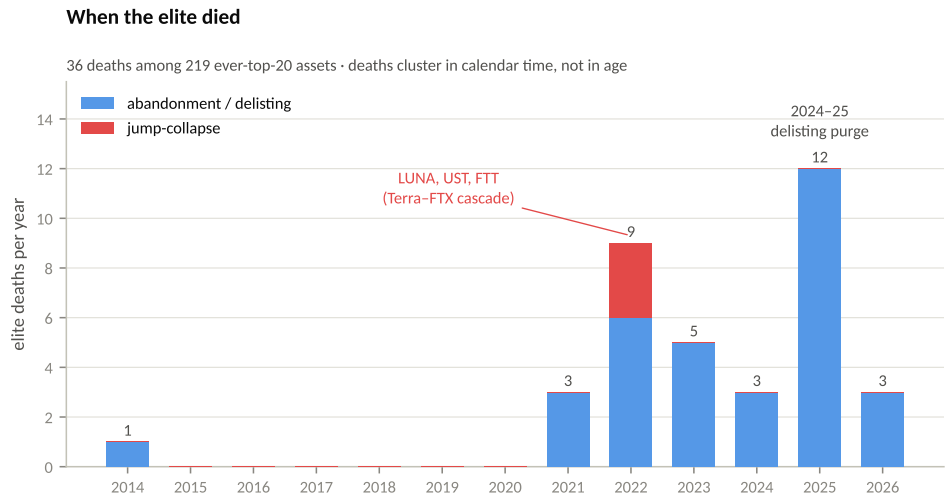


Figure 4: When the elite died: 36 deaths among 219 ever-top-20 assets, by year and type.

given after a strong run is not the long-run expectation the condition prices. The paper therefore does not calibrate risk aversion to the surveys. It uses the household-finance range of Section 2, $\gamma \in [1.7, 2.8]$, and treats the stated expectations as a lower reading: what holders say they expect is at or below what their positions require.

Source 3: option prices. Options give a holder-free reading. A classic result (Martin 2017) says the risk-neutral variance priced into options (the variance computed from option prices rather than from history) is a lower bound on the expected excess return demanded by an investor fully invested in the asset; in equation (1) "$\gamma w = 1$" describes a fully invested log-utility investor, whose required excess return is exactly the return variance, and Martin's result is that option prices put a floor at that variance. Deribit-based estimates for 2017–2022 put that bound at **66–84% a year**, with Bitcoin implied volatility then running at 80–90% (Almeida, Grith, Miftachov, and Wang 2024; Deribit options, July 2017–December 2022). This is the concentration mechanism priced by the market itself, in the era when crypto markets were documentably segmented from mainstream capital, with large and persistent cross-market price gaps (Makarov and Schoar 2020), and the marginal trader really was close to all-in. Today Deribit's DVOL index (the 30-day option-implied volatility, a VIX for Bitcoin) has run between about 40 and 55, so the same all-in bound is 0.40^2–$0.55^2 \approx$ **16–30 points of excess return**, the requirement of a holder with all her wealth in the asset; the actual marginal holder, who in the ETF era is no longer all-in but holds w of 8–19%, requires $\gamma\sigma^2 w \approx$ **2–9 points** for concentration, on top of the risk-free rate and the market term.

Putting the 2026 number together. Risk-free 5.0%; market term 6–8 points (beta 1.5–2.0 on a 4% equity premium: the since-2016 and four-year estimates and the range of the rolling beta since 2022, Appendix A); concentration 2–9 points ($\gamma \in [1.7, 2.8]$, $\sigma = 42\%$, $w \in [0.08, 0.19]$); mortality 0.3–3 points. Summing the extremes gives 13.7% to 25.4% a year, an *envelope* of 14–25%. But an envelope puts every uncertain input at its extreme at once, and the inputs are independent. Treating each as uniformly uncertain over its stated range and drawing them independently (Appendix D), the required return has a median of **about 19% a year**, a central 68% interval of 17–21%, and a **90% interval of 16–22%**. The market term is now the largest single component (about 7 points), followed by concentration (about 5); the holder weight still con-

tributes most of the width. This is an arithmetic drift; the compound return it implies is lower by about half the variance, roughly 9 points at today's volatility, so about 10% a year. The age rule, for contrast, says 32% with a precision that is apparent rather than real: a one-parameter fit returns a point by construction. And the realized five-year outcome will carry, on top of the parameter uncertainty described here, diffusive noise of roughly ± 19 points on the compound return at one standard deviation, which no forecast of a 42%-volatility asset can remove.

6. THE REQUIRED RETURN THROUGH TIME, AGAINST WHAT BITCOIN DELIVERED

A cost of capital earns its name if, over time, the asset delivers about what it says. This section runs that test: the required return is evaluated at each date from 2012 from that date's inputs and compared with what Bitcoin delivered over the following five years.

Matching units. The comparison has to be made in the units of equation (1). The required return is an arithmetic drift, the expected rate of price growth. The compound annual growth rate (CAGR) an investor realizes is lower than the drift by about half the variance, and at Bitcoin's historical volatility of 60–150% that is 20 to 100 points a year. Comparing a required drift with a realized CAGR would understate what Bitcoin delivered by that much. We therefore measure the *realized drift*: over each five-year window, the mean log return plus half the realized variance, annualized. The CAGR is reported alongside for reference. Five years is the horizon of the companion paper's backtest and long enough to span a four-year market cycle.

The required return at each date, $\mu^{req}(t)$, is equation (1) evaluated from the ingredients measured at that date: trailing volatility from the price series; the holder weight w bounded between a market-clearing back-cast and the all-in trader the option bounds describe; the risk-free rate of the date; the market term from a rolling four-year beta (Appendix A); and the mortality term held at its stated range throughout, since the elite hazard is one average over 2014–2026. The back-cast generates the whole pre-2021 lower band, so it deserves plain words; Appendix B shows each step graphically. Market clearing requires that Bitcoin's market capitalization equal the wealth its holders have placed in it: cap $= \bar{w}\times$ (participating wealth). We anchor participating wealth in 2021Q3, the one date where both sides are measured (capitalization divided by the surveyed 19%), and move it backward and forward in proportion to the number of participants times average wealth per participant. The participant count is the share of US adults holding crypto in the household surveys of Section 3 (0.4% in 2014, 1.2% in 2018, 11% in 2021Q3, 12–13% since), interpolated log-linearly between survey dates, with the count of addresses holding a balance used only before the first survey; wealth per participant is a ramp from a quarter of its 2021 level in 2012, when holders were students and hobbyists, to 1.4 times by 2026, when they include institutions. From 2024 the spot ETFs and public-company treasuries add a second channel of holders who are not "crypto owners" in any household survey and hold small weights; the lower edge of the band counts their roughly 2.5 million BTC (mid-2026) as held at 1.5% weights, the upper edge does not count them. Dividing capitalization by that path gives the implied average holder weight at each date: about 3% in 2013, 17% in 2017, 18% in 2021 (the anchor), and in 2026 between 8% with the institutional channel and 21% without it. Read literally, the low early values say that *if* early Bitcoin wealth had been spread across survey-style households, each held a small position and required little; the all-in alternative reads the same capitalization as concentrated in few fully committed hands who required a great deal. The band is exactly the interval between those readings. Table 4 reports

the band at two-year intervals; Figure 5 shows the full monthly series.

Table 4: Required return versus realized drift, by starting date.

Start	Volatility	Beta	Required band	Midpoint	Realized drift	Realized CAGR	Age rule
2013	151%	3.0	26–231%	128%	144%	135%	132%
2015	115%	2.9	21–135%	78%	99%	103%	90%
2017	80%	2.2	28–67%	47%	76%	51%	68%
2019	90%	0.7	38–82%	60%	58%	42%	55%
2021Q3	77%	1.3	23–47%	35%	27%	13%	45%
2023	66%	2.1	18–29%	23%	–	–	40%
2026	42%	1.75	14–27%	20%	–	–	33%

Rows are mid-year except 2021Q3, the survey date, whose realized figures run to May 2026 (4.7 years); other realized figures cover the following five years. The required band runs from the survey-anchored back-cast of the holder weight (lower edge) to the all-in option bound (upper edge). Volatility and beta are trailing estimates at the start date. For 2026 the propagated estimate of Section 5 is 19% (90% interval 16–22%). Age rule: five-year form.

The test. Over the 113 monthly windows starting January 2012 to May 2021, the realized drift exceeded the midpoint of the required band by 5 points a year on average, against levels of 30 to 150%: on average, Bitcoin delivered what the condition says its holders required. Since the 2021 survey the realized drift has been 27%, inside the 23–47% required at that date. The lower edge of the band alone (survey-type holders with modest weights) would leave 60 points a year unexplained, and the upper edge alone (the all-in holder) would overstate by 49; the data cannot say which holder was marginal at each date, and the midpoint is an equal-weight reading of the two, not an estimate. Realized drift falls inside the band in 48% of windows.

What the test does not show. The required return gets the level right on average, not the timing. Its mean absolute miss is 51 points a year, against 21 for the age rule, and its correlation with subsequent realized drift is 0.14, against 0.83. The reason is visible in Figure 5: the band is built from trailing volatility and from a holder weight with market capitalization in its numerator, so it is highest at market peaks, exactly when subsequent returns are lowest (Appendix D reports the regression). A date-by-date equilibrium restriction is not a timing signal, and the paper does not offer it as one. The age rule tracks the path better because it is a smooth curve fitted to that path; what it lacks is a reason, and so a way to know when it will stop.

Why the bands are wide early. The uncertainty in 2013–2017 is *who* the marginal holder was. Under the survey-anchored back-cast, participants were few but each held a modest weight, and the required return was 20–30%. Under the all-in reading, which the 2017–22 option bounds support for the segmented era, required returns were in the tens-to-hundreds. (In the language of the companion scarcity framework (Papadogiannis 2025), where network value is desirability times scarcity amplified by adoption, the lower reading attributes the early returns to unanticipated growth of that value function; the upper reading attributes them to compensation for concentrated risk.) Realized drift sits between the two readings, nearer the middle than either edge, and for the paper's purpose the question does not need to be settled: on *either* reading the early engine (concentration, volatility, segmentation) has measurably shut down.

Which term carried the requirement, and when. Figure 6 takes the midpoint of the band and splits it into the four pieces of equation (1), in levels and as shares. In 2013 the concentration term, $\gamma\sigma^2 w$, was about 88% of a requirement near 128% a year. It still supplied three-quarters in 2021. After the ETF-era dilution of the holder base and the fall in volatility the composition changed: in 2026 the market term is about 35% of

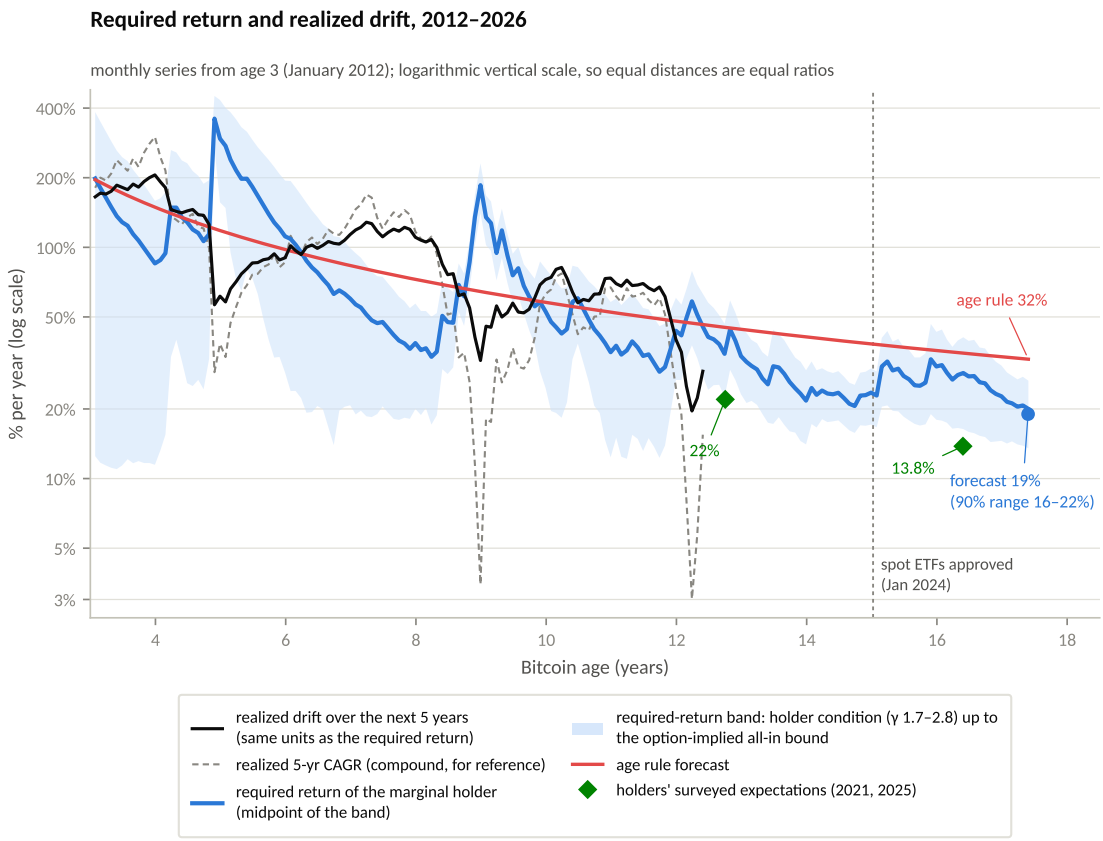


Figure 5: Required return and realized drift, on a logarithmic vertical scale so that equal distances are equal ratios. Series begin at age 3 (January 2012), the start of the companion paper's backtest sample and the first date with enough price history for a trailing volatility estimate; the realized curves end at age 12.4 because each point covers the following five years. The solid black line is the realized drift, in the units of the required return; the dashed grey line is the realized CAGR. Green diamonds mark holders' surveyed expectations; the dotted line marks spot-ETF approval (January 2024); the right edge marks the 19% forecast against the age rule's 32%.

the midpoint requirement, concentration 33%, the risk-free rate 24%, and mortality 8%. Two things follow. The history of Bitcoin's required return is mostly the history of one term, which is why Section 2 could build the equation around it. And the asset has changed character: its required return is now mostly the price of ordinary market risk and the risk-free rate, the profile of a high-beta risk asset, with a concentration premium that is still material but no longer dominant. (The midpoint path ends at 21% rather than the 19% of Section 5 because the midpoint weights the band's corners equally, which the propagation of Appendix D does not.)

What the age rule is, on this account. Both the required return and the age rule fell from over 100% to tens of percent across the sample, and a one-parameter power law in age is an excellent smoother of a declining curve. The age rule is best read as a reduced form of the required return: right on average over a period in which the inputs fell roughly as one over age, with no structure to say whether they will keep doing so. They have not. Volatility, the holder weight and the mortality term do not decline with age as such; they declined with integration, and the market term has risen with it. The age rule's 33% for 2026 sits

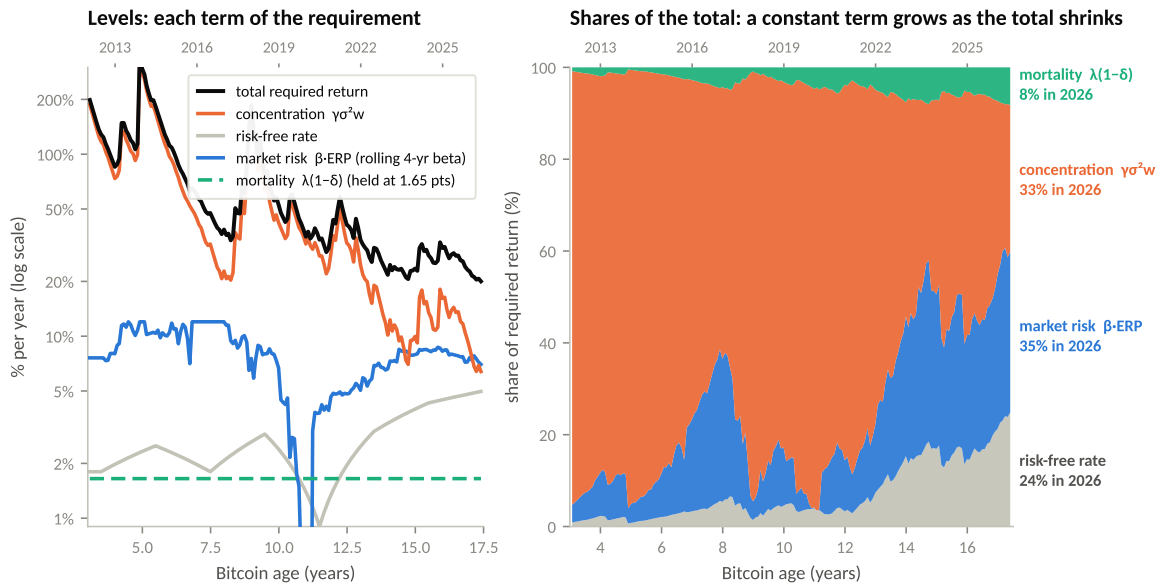


Figure 6: The four terms of equation (1) over Bitcoin's life, from the midpoint of the required-return band. Left: the level of each term and of their sum, in percent a year on a logarithmic scale. Right: each term's share of the total. The market term uses the rolling four-year beta of Appendix A. The mortality term (dashed) is held constant at the midpoint of its stated range, 1.65 points, so its share rises as the total falls. Concentration is the remainder, which includes the all-in option bound at the band's upper edge.

above the top of the required band (27%), and the companion backtest's own out-of-sample record is in the same direction: in windows beginning after the power law's 2019 publication, Bitcoin fell short of the age rule by 5–8 points a year.

Premium or belief? The concentration term is mechanical: a holder with a large fraction of her wealth in one volatile asset bears a great deal of undiversified risk, and $\gamma\sigma^2 w$ is the return that makes keeping it rational, whatever put her there. In Merton's (1987) story only a few investors can or will hold the asset, so each holds a lot and the price must pay them a premium, which arrives on average. In the other story investors pile in because they expect a great deal; their weights are the sizing rule of Section 9 run on inflated beliefs, and because the price has already risen to meet the belief, the return does not arrive. From the weight alone the two are indistinguishable. What separates them is whether the return arrived, and over 2012–2021 it did, on average and in the units of the condition. That is evidence for the premium reading over the sample as a whole, though not at every date: the required return was highest at the 2013, 2017 and 2021 peaks, and the returns that followed those dates were below it, which is what belief-inflated weights at peaks would produce.

The cross-section agrees from the side. If required returns rise with volatility, concentration and death risk, the typical altcoin (more volatile, more concentrated, higher λ than Bitcoin) required more than Bitcoin did. It delivered less than nothing in compound terms: across the significant-asset cohorts, the *median* coin born 2015–2020 shows a negative compound return at every horizon, while cohort *means* stay large on the back of a few winners, exactly the lottery shape that a 10–17% annual death rate plus a handful of extreme

survivors produces. Within Bitcoin's own 2010–14 birth cohort at age ten, the mean coin compounded at about +167% a year, the median at +50%, and Bitcoin at +233%. Bitcoin is the asset for which realized and required coincide on average; it is also the winner of its cohort, and a required return validated on the winner is validated on a selected path, a caveat Section 10 repeats.

7. THE FORECAST, AND WHAT WOULD PROVE IT WRONG

A required return is a statement about risk; a forecast is a statement about the price process. The bridge from one to the other is an equilibrium condition, which we state as our hypothesis and own: *Bitcoin is priced in equilibrium by its marginal holder*, with no systematic mispricing in it. Section 6 gives the hypothesis some standing, since the required return has been right on average for fourteen years, and the spot ETFs, the dilution of the holder base, and the compression of the option bounds all point to a market more integrated now than at any earlier date. Under that hypothesis, and only under it, the expected return equals the required return of the marginal-holder condition, every ingredient of which is observable:

If Bitcoin is priced in equilibrium, its expected return for September 2026 to September 2031 is about 19% a year in drift, with a 90% range of 16–22% (full envelope 14–25%); in compound terms about 10% a year (7–13%; envelope 5–16%). The age rule says 32% (22–28% compounded, on the companion paper's conversion). The two forecasts differ enough to be separated by a single five-year window, and we commit to ours in print.

To keep the claim honest, here is what each outcome would mean:

- **BTC compounds (CAGR) at $\geq\sim$25% a year to 2031.** The age rule is vindicated and this paper's account is wrong in an instructive way: either a required-return component we measure as small is in fact large (which the options market would have to show), or Bitcoin remains systematically underpriced, which is the age rule's implicit claim.
- **BTC compounds between the mid single digits and the high teens.** The account stands; the age rule's first structural break is confirmed, for the reason stated here.
- **BTC compounds near zero or below.** Within the band's logic this is a bad draw of the (large) diffusive risk, not a refutation. But if it comes with *rising* option-implied crash intensity well above the graveyard's 0.3–3%, the mortality term was underestimated.

Two notes on the test itself. *Units:* the 19% is an arithmetic drift; at today's $\sigma \approx 42$–45% the volatility drag is roughly $\sigma^2/2 \approx 9$ points, which gives the compound figures above; the realized outcome should be graded in the same units as the forecast it is compared with, as Section 6 does. *Asymmetry:* as the three outcomes make explicit, the forecast is sharply falsifiable upward but only conditionally falsifiable downward. A one-and-a-half-sided test is the price of forecasting a 45%-volatility asset, and we state it rather than hide it.

What about the halving cycle? Practitioners will ask where Bitcoin's four-year "halving cycle," the pattern of peaks in 2013, 2017, and 2021 that followed the supply halvings of 2012, 2016, and 2020, fits in this account. Four points. First, the halving changes supply, and supply does not appear in equation (1): cutting issuance from 0.8% to 0.4% of the outstanding stock a year alters none of the risks a holder is paid for, so the required return is unaffected by the schedule; supply matters for the price level, which this paper

does not model. Second, the schedule is written in the code and known years ahead, and in an equilibrium a known future supply change is in today's price and cannot carry a predictable excess return around its date. A reliable, tradeable cycle would therefore be a systematic mispricing, and the equilibrium hypothesis above says there is none left; the forecast is, among other things, a bet that the cycle is not a tradeable regularity going forward. Third, the past cycles were real and are in Figure 5: the five-year windows starting at the 2013, 2017, and 2021 peaks are the troughs of the realized curve, those starting in the 2015 and 2019 troughs its crests. In this paper's language the cycle is realized drift swinging around the required return, and the condition says nothing about its timing, only that the swings average out in equilibrium, as they did over 2012–2021; the regressions of Appendix D, where the concentration quantity spikes at market peaks and predicts low forward returns, are the bust leg of that cycle. Fourth, the paper predicts that the cycle's amplitude and predictive content should fade, because the mechanism that produced it, a small concentrated holder base at 80–150% volatility, is the mechanism documented compressing here and in Figure 7, and because an integrated market arbitrages a four-year calendar. The forecast window, September 2026 to 2031, straddles the 2028 halving and whatever peak or trough follows it, so the cycle phase is partly averaged out of the test, which is one more reason for the five-year horizon.

Two auxiliary predictions sharpen the mechanism. First, the required return should fall fastest where volatility and holder concentration fall fastest: Ether post-ETF should show the same compression; a still-segmented, still-concentrated large asset should not. Figure 7 shows the first check already passes: Ether's volatility and forward returns compress alongside Bitcoin's into the ETF era. Second, option-implied crash intensity should remain in the low single digits, consistent with the measured elite jump hazard; a reading persistently above $\sim$10% a year would push our band up and signal that the market disagrees with the graveyard, whether about contagion, regulation, or forward-looking tail risks such as a quantum break of Bitcoin's signature scheme (Appendix C.4), which realized mortality cannot yet contain.

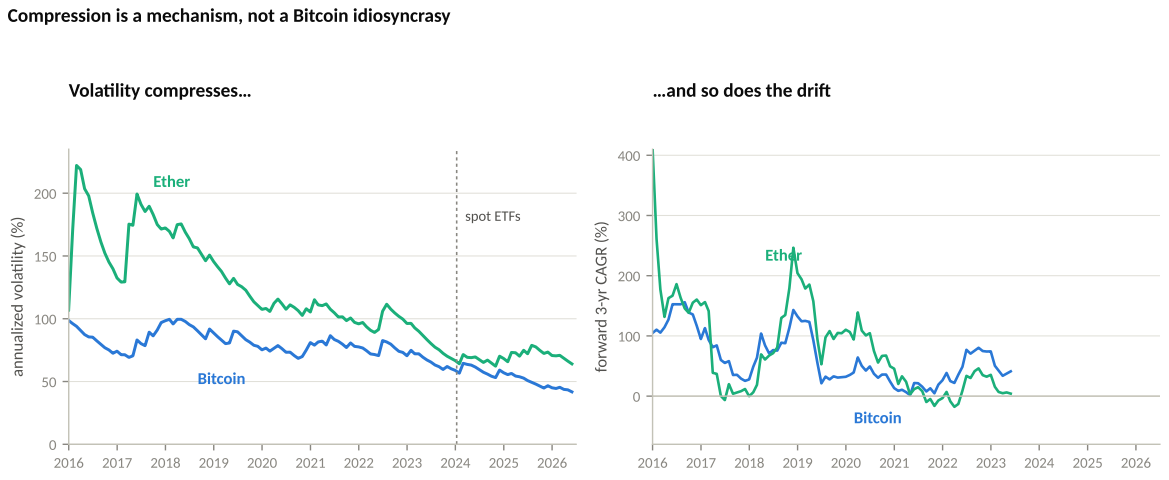


Figure 7: The compression mechanism is not Bitcoin-specific. Left: EWMA annualized volatility for Bitcoin and Ether, with spot-ETF approval dates marked. Right: realized forward three-year returns for both assets.

8. THE SAME PRICING FOR ETHER AND OTHER CRYPTOASSETS

Nothing in equation (1) is specific to Bitcoin. The condition prices any asset held by a concentrated investor who faces a risk of total loss; what changes from one cryptoasset to the next is the value of the inputs. The graveyard was built from the whole asset class, so it already supplies the death-risk input by size class, and Bitcoin's volatility and holder-weight measurements have natural counterparts. Table 5 walks the condition down the size ladder.

Table 5: The marginal-holder condition across the cryptoasset size ladder. Required returns per year, as arithmetic drifts. The central estimate and 90% interval come from drawing each input independently and uniformly over its stated range (Appendix D); the envelope sums the extremes. Risk-free 5%, equity premium 4%, $\gamma \in [1.7, 2.8]$ throughout. Beta is 1.5–2.0 for Bitcoin and 2.0–2.7 for Ether (since-2016 and four-year estimates); Ether's range is assumed for the altcoins. Altcoin holder weights are illustrative, not surveyed; small-coin volatility is taken as 100–130%.

	Bitcoin	Ether	Large altcoin	Small coin
Volatility σ, 2026	42%	64%	∼80%	100–130%
Market beta	1.5–2.0	2.0–2.7	2.0–2.7	2.0–2.7
Holder weight w	0.08–0.19	0.08–0.19	0.10–0.30	0.10–0.30
Death hazard, per year	0.3% jump	0.3% jump	3.2% all-cause	17.4% all-cause
Mortality term	0.3–3 pts	0.3–3 pts	1–3 pts	16–17 pts
Required return, central	**19%**	**28%**	**45%**	**88%**
90% interval	16–22%	23–35%	31–62%	60–131%
envelope	14–25%	19–41%	25–73%	46–175%
Delivered, drift (compound)	27% (13%)	19% (−7%)	∼25% (−7%)	not observable

Delivered: Bitcoin and Ether, realized drift and compound annual return from the 2021Q3 survey date to May 2026 (Section 6); large altcoins, the median compound return of the significant-asset cohorts born 2015–2020, measured five to seven years after birth (−6.5% to −8.7% a year), with the drift obtained by adding half the variance at 80% volatility. Small coins have no usable price history in the data; what is observable is that half of the smallest quartile is dead within 3.3 years (Figure 3). Hazards are the graveyard's size-quartile and elite estimates (Section 4).

Ether. Ether has been a top-five asset for a decade, so its death risk sits in the same elite class as Bitcoin's and the mortality term is the same 0.3–3 points. It differs in two inputs: volatility, 64% against 42% today, and market beta, 2.0–2.7 against 1.5–2.0. On the holder weight we make one assumption: surveys report a household's crypto holdings as a block, Ether's monthly returns have a correlation with Bitcoin's of about 0.8, and so the share that matters for a holder's risk is her crypto share, the same 8–19%. The concentration term scales with the ratio of variances, $(0.64/0.42)^2 \approx 2.3$, and Ether's required return comes out at about 28% a year (90% interval 23–35%; envelope 19–41%). From the 2021Q3 survey date to May 2026 Ether's realized drift was about 19% (a compound return of −7%; the gap is the volatility drag), below the interval: unlike Bitcoin, Ether has delivered less than the condition requires over this window. Figure 7 shows Ether's volatility compressing alongside Bitcoin's; the condition says its required return has compressed with it.

Large altcoins. For the surviving assets in the largest size quartile outside the top two, volatility is about 80% and the all-cause death rate is 3.2% a year, mostly slow fades in which holders recover part of their value, so a mortality term of 1–3 points. The holder weight is not surveyed for individual coins; concentration among committed communities is plausibly higher than for Bitcoin, and we show an illustrative 0.10–0.30. The required return lands at about 45% a year (90% interval 31–62%; envelope 25–73%). The significant-asset cohorts of Section 6, which are the surviving large altcoins, delivered median compound returns of −7% to −9% a year five to seven years after birth, a drift of roughly 25% once the volatility drag is added back: the typical large altcoin has fallen short of what its risk required by some 20 points a year in

drift, and lost money in compound terms.

Small coins. At the bottom of the ladder the mortality term stops being a footnote. The smallest quartile dies at 17.4% a year, with recovery close to nothing, so the death risk alone adds about 16 points to the required return before any compensation for volatility of 100% or more. The condition demands about 88% a year (90% interval 60–131%), and 46% at the very least. What was delivered is not observable in the price data, which the free tier does not carry for small coins; what is observable is that half of them are dead within 3.3 years of listing (Figure 3), and that even the surviving significant altcoins, a far stronger group, fell short. The pattern is the one documented for lottery-like stocks, where investors accept low average returns for a small chance of an extreme payoff (Barberis and Huang 2008); the graveyard prices that preference in this asset class, and for the half that die within three years the shortfall is the whole stake.

The altcoin holder weights and betas are illustrative and the realized comparisons cover only surviving assets with price history, which flatters them (Section 10); the direction of the result does not depend on either. As one moves down the size ladder the required return rises steeply, because volatility and death risk both rise, while realized returns fall short by more. Bitcoin is the one cryptoasset for which the two sides of the condition have met, which is a statement about Bitcoin's holder base and survival record, not about the condition.

9. FOR THE ALLOCATOR

The required return translates directly into practice. This section states the hurdle a diversified allocator should apply, inverts the condition into a sizing rule, places the result in a Black–Litterman frame, and then asks the question the numbers invite: whether, and for whom, Bitcoin is worth holding at all.

The hurdle. A diversified allocator should require of Bitcoin what its contribution to her portfolio's risk costs her: the risk-free rate plus beta times the equity premium, about **11–13% today**, plus some or all of the mortality term, 0.3–3 points, to the degree she judges crypto mortality systematic rather than diversifiable. The 2022 clustering of elite deaths (Terra, Celsius, FTX within seven months) argues it is partly systematic; we would charge 1–2 points. Call the hurdle 12–15%. This is not a forecast of what Bitcoin will do; it is the bar it must clear to belong in her portfolio at more than market weight. It is twice what it would have been before 2020, because Bitcoin now moves with equities.

The sizing rule. The marginal-holder condition, inverted, sizes the position. If an investor's own view of Bitcoin's expected return is $\hat{\mu}$, her risk aversion γ, and Bitcoin's volatility σ, her weight should be

$$w^* = \frac{\hat{\mu} - r_f - \beta\,\mathrm{ERP} - \lambda(1-\delta)}{\gamma\sigma^2}. \tag{2}$$

Worked example: an investor with $\gamma = 3$ who believes our central forecast, $\hat{\mu} = 19\%$, at today's $\sigma = 42\%$ ($\sigma^2 \approx 0.18$) and a beta of 1.75: $w^* = (19 - 5 - 7 - 0.3)/(3 \times 0.18) \approx$ **12% of financial wealth**. That is aggressive by institutional standards, and about two-thirds of the 19% the average 2021 holder actually ran at far higher volatility. An investor who believes the age rule's 32% gets $w^* \approx 36\%$; the distance between those two positions is the practical cost of getting this paper's question wrong.

The Black–Litterman translation. In a Black–Litterman portfolio, the tilt away from market weight is proportional to the gap between your expected return and the market-implied one (Black and Litterman 1992). This paper reduces that gap to observable disagreements: your volatility path, your view of holder-

base integration (the path of w), and your mortality estimate against the market's option-implied one. An allocator with no disagreement on those three has no basis for holding Bitcoin away from its market weight, currently well under 1% of global financial wealth, however glorious its history.

Is it worth holding at all? The results invite the question. For the smaller cryptoassets the answer is no: required returns of 45–88% against delivered returns that fall short or cannot be observed mean their holders have been underpaid on average, and the only coherent account of their demand is the lottery preference of Section 8. For Bitcoin the answer depends on who is asking. The 19% is what its concentrated holders require, and in equilibrium it is what they get, so for them it carries no alpha; "fairly priced" means exactly that. A diversified investor is in a different position. She does not bear the concentration risk those holders are paid for, and at a market weight near 1% she barely bears the death risk either: a jump to zero would cost her 1% of wealth. Her hurdle is the 12–15% above, and the asset offers about 19%. The 4–7 point difference is a rent, the premium a still-concentrated holder base demands, available to anyone able to hold the asset as a small part of a diversified portfolio; that is Merton's (1987) original observation about limited participation, and it is why the sizing rule above returns a weight of about 12% for a $\gamma = 3$ investor who accepts the central forecast, rather than zero. The rent is what remains of the concentration term, and it shrinks as the holder base broadens.

Against equities, standing alone. The comparison can be made with the paper's own inputs. Take the S&P 500, the index against which Bitcoin's beta is measured. Its expected excess return is the equity premium used throughout, 4% (6% in the high case); its volatility is 16–20% a year over the long historical record (Dimson, Marsh, and Staunton 2002) and 16.5% in the current J.P. Morgan assumptions (J.P. Morgan Asset Management 2026). The ratio of excess return to volatility, the Sharpe ratio (Sharpe 1994), is therefore about 0.20–0.25 and up to about 0.35 in the high case. Bitcoin at a 19% required return, a 5% risk-free rate, and 42% volatility gives $(19-5)/42 \approx 0.33$, and about 0.29 once the mortality term, which is compensation for a loss and not a return, is taken out. If the required return is paid, Bitcoin offers somewhat more reward per unit of risk than equities, and the excess is the concentration premium. The choice of index does not change the comparison: a global all-cap index (FTSE Global All Cap or MSCI ACWI IMI, some 10,000 stocks across developed and emerging markets) has nearly the same figures as US large caps, since the J.P. Morgan assumptions give world equity an arithmetic return of 8.3% at 16.8% volatility against 7.9% at 16.5%. A reader who prefers the historical US premium of 7–8 points over bills (the familiar 10% nominal return of the last century) should raise both sides: equities' ratio rises to about 0.45, and Bitcoin's market term doubles with the premium, taking its required return to about 26% and its ratio to about 0.50. What has changed is the diversification argument. Bitcoin's correlation with equities, 0.1–0.25 before 2020, has been about 0.5 since 2022 (Appendix A; Copestake, Furceri, and Terracciano 2026), so more than a third of its required return is now payment for a risk the equity investor already holds. What this paper removes is not the case for a small Bitcoin position but the case built on 32% and on zero correlation. At a central 19% in drift, about 10% compounded, with a beta near 1.75, it is a high-beta risk asset with a shrinking concentration premium.

In Sharpe units. Dividing equation (1) by σ turns the required return into a required reward per unit of risk, and because $\beta = \rho\,\sigma/\sigma_m$ the market term simplifies:

$$\frac{\mu - r_f}{\sigma} \;=\; \rho\,\mathrm{SR}_m \;+\; \gamma\sigma\,w \;+\; \frac{\lambda(1-\delta)}{\sigma}, \tag{3}$$

where ρ is the asset's correlation with the market and SR_m the market's Sharpe ratio. For a fully diversified holder of an asset that cannot die the right side collapses to $\rho\,\mathrm{SR}_m$, the CAPM in Sharpe units: no asset can offer more reward per unit of risk than the market times its correlation with it. The concentration term, $\gamma\sigma w$, is what a holder adds when she owns a lot; σw is the position's contribution to her portfolio volatility, so Sharpe stops being scale-free. The mortality term shrinks with volatility, because a fixed expected loss is a smaller share of a larger risk budget. More generally, at any holder's optimum an asset's Sharpe ratio equals its correlation with her own portfolio times that portfolio's Sharpe ratio; concentration is simply what makes the correlation high. The numbers are telling. Today the market term (7 points over 42%, a correlation of about 0.5 at a market Sharpe ratio near 0.25–0.3) is about 0.15, the concentration term ($2.25 \times 0.42 \times 0.13$) about 0.13, and mortality (1.65 points over 42%) about 0.04: a required Sharpe ratio of about 0.32, the figure reached above by division. In 2013 the band midpoint implies about 0.8, and above 1 at its all-in edge, a reward per unit of risk that no asset class delivers for long and Bitcoin did not. Once holders are diluted toward market weight ($w \to 0$) the requirement falls to $\rho\,\mathrm{SR}_m$ plus the mortality term, about 0.19 at today's volatility: an asset 0.5-correlated with equities should, in an integrated market, earn about half the market's reward per unit of risk plus a few points for death risk. Bitcoin's current Sharpe ratio exceeds that only because its holders are still concentrated, and integration should push it down toward equities' own. The same form answers the allocator's question directly: an asset improves a diversified portfolio's Sharpe ratio if and only if its own Sharpe ratio exceeds ρ times the portfolio's, here about 0.32 against about 0.13, so a small position adds today; the margin is the limited-participation rent plus the mortality allowance, and the rent vanishes as $w \to 0$ by construction. For gold, with $\rho \approx 0$ and a Sharpe ratio of a few hundredths, the improvement is negligible on expected-return grounds; whatever case gold has rests on tail behavior this condition does not price.

Appendix E applies the same condition to gold, where it reproduces a known result (a required return near the risk-free rate; Barro and Misra 2016), and spells out what "digital gold" would mean in return terms: about 14% a year at gold's volatility and holder dilution, because Bitcoin would still carry its beta and its death risk, and gold's 5–6% only if it shed both.

10. LIMITATIONS AND THE MEASUREMENT AGENDA

This is a measurement paper, so its limits are measurement limits; we list them in the order a critical reader should weigh them.

1. **The holder weight before 2021 is bounded, not measured.** The portfolio-share surveys begin in 2021; everything earlier rests on market-clearing back-casts and option bounds, and the 2026 lower bound on an assumed weight for ETF and treasury investors (Appendix B). The paper's historical bands are honest about this width, and its forward claim does not depend on the early years. The single most valuable upgrade would be a measured w path: entity-adjusted on-chain concentration joined to holder-wealth estimates, the 2025 portfolio-share figures in the Candia et al. online appendix (not yet published in text at this writing), and the ETF share of total holdings.

2. **Coverage-onset selection in the graveyard.** Assets enter the panel only after becoming significant, so young-age hazards are understated. This biases the data *toward* Lindy, which makes the rejection

conservative, and it leaves the elite estimates untouched; but population hazard *levels* at ages under two should be read as floors.

3. **Death timing and classification.** Feed and volume deaths lag economic death by months; four elite "deaths" are contested zombies (chains that operate far below their manic highs); one (ICP) is an artifact of a listing-day price spike. All classifications, dates, recovery fractions, and confidence flags are in Appendix C, and the headline elite hazard moves only from 3.5% to 3.15% when the contested cases are dropped.

4. **Bitcoin's own existential era is unmeasurable.** No cross-section existed in 2010–2014 to price the risk that Bitcoin itself was a dead end. Our early-period required-return bands are therefore wide by construction, and nothing in the paper's forward claim rests on resolving them.

5. **Survey-to-wealth pairing.** The participation series is US-based and the wealth aggregates global; re-running the clearing calculations with US household wealth against US participation moves the bands by amounts well inside their width (Appendix D).

6. **The holder weight is endogenous to beliefs.** A weight chosen on over-optimistic expectations satisfies the same first-order condition as one forced by segmentation, so the concentration term recovers the holder's subjective hurdle; whether that hurdle was a fair premium is adjudicated by realized returns (Section 6) and by the option-implied bound, not by the equation. The forecast assumes that the ETF-era weights are participation-driven; that assumption is what the 2031 test checks.

7. **The condition is complete only for the holder it describes.** Liquidity, jump-risk and ambiguity premia (upward) and convenience yield, skewness preference and disagreement (downward) are not priced separately; the option-implied bound and the average agreement with realized drift leave little room for a large net residual, but not none (Section 2, fourth clarification).

8. **The validation is on a survivor, and on average only.** Realized and required drift agree on average over 2012–2021, but Bitcoin is the winner of its cohort, the band is wide, the midpoint is a convention rather than an estimate, and the required return does not track the timing of returns (Section 6). The 2025 survey answer is below the model (Section 5).

9. **The market beta is unstable.** The rolling four-year beta has ranged from about zero to 3 and the correlation from zero to 0.55 (Appendix A); the forecast uses 1.5–2.0, and a return to pre-2020 correlations would lower the required return by several points.

10. **Few independent windows.** Five-year returns from a 16-year price history contain roughly three independent observations. That is why the paper leans on levels at measured dates, cross-sectional mortality, and two survey waves rather than on time-series regressions, and why its forecast is phrased so that a single new window can grade it.

11. **Illustrative inputs for the other assets.** The altcoin holder weights and betas of Section 8 and gold's holder weight in Appendix E are assumed ranges, not survey measurements, and gold's mortality allowance is a judgment; the required returns built on them are directional, and the Bitcoin estimate, which rests on measured inputs, does not depend on them.

11. CONCLUSION

The riddle was that Bitcoin seemed to carry three expected returns at once: 32% from its own tested history, 11–16% from the CAPM, 14–22% from its holders' mouths. The resolution begins with classification. Only the first was ever an expected return, a property of the price process. The CAPM range is a normative required return, the hurdle of a diversified investor. The third, what the people who hold Bitcoin require, had no model behind it, and this paper supplies one: Merton's (1987) limited-participation pricing with measured inputs, a market term from a rolling beta, a concentration term from surveyed portfolios, and a mortality term from a census of 1,815 cryptoassets. The graveyard shows mortality was never the main story (hazard rises with age; the top-20 jump hazard is 0.3% a year). Concentration was: it supplied most of a required return that ran above 100% a year in Bitcoin's early life. And when that required return is compared with what Bitcoin delivered, in matching units, the two agree on average over fourteen years. Bitcoin's history looks less like an anomaly than like a very large risk premium, paid.

What remains, if the asset is priced in equilibrium, is a required return built from observable things: 40-some percent volatility, a beta to equities near 1.75, a holder base diluting toward market weight, a small and datable mortality risk, and a risk-free rate. Those ingredients say about 19% a year in drift for 2026–2031, with a 90% range of 16–22%, or about 10% compounded; the same condition, fed gold's inputs (Appendix E), returns the risk-free rate plus half a point. The age rule's 32% sits above the band, and the framework says why it should fail: it extrapolates a decline that was driven by integration, not by age, and integration has now raised one term while lowering the others. If Bitcoin instead repeats its age-rule 32%, then either it is still underpriced or a risk we measured as small is large, and the framework says which observables separate the two.

For practitioners. Four takeaways. Use about 19% a year (16–22%) as Bitcoin's expected arithmetic return in capital-market assumptions, about 10% compounded, not the 32% its history extrapolates. Do not treat it as uncorrelated: its beta to equities is about 1.75 and its correlation about 0.5, so a diversified investor's hurdle is 12–15%, and the 4–7 points above that are a limited-participation premium that will shrink as ETFs broaden the holder base. Size the position with the condition inverted (equation 3): about 12% of financial wealth for an investor with risk aversion of 3 who accepts the central forecast, and proportionally less for higher risk aversion or a lower view. And charge for mortality explicitly, 0.3–3 points, with option-implied crash intensity as the running check on whether the market has begun to price a larger hazard. The measurement agenda of Section 10, above all a measured path for the holder weight, is where the estimate can be sharpened; the forecast itself is graded in September 2031.

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Data and code availability: the full replication package (panel-construction and survival scripts, the curated asset-level panel, all result tables, and every exhibit) is maintained at `https://github.com/apapadogiannis/bitcoin-required-return`. The pipeline ends with an audit script that recomputes and checks every quantitative claim in this paper. Underlying market data are the public Coin Metrics community dataset (snapshot ending May 24, 2026), which the package downloads from its source rather than redistributing.

A. MARKET RISK

A.1 Estimation. The market term of equation (1) is Bitcoin's beta to the equity market times the equity premium. Beta is estimated by regressing Bitcoin's monthly log return on that of the S&P 500, both measured month-end to month-end; the index is represented by the total return of the Vanguard S&P 500 ETF (VOO, adjusted close), whose month-end ratio to the index is constant to the third decimal over the sample. Over the full windows the estimates are 1.75 for the last four years (correlation 0.50), 1.5 since 2016 (0.32), and 1.8 since 2012 (0.29). The time path used in Section 6 is the rolling four-year beta (an expanding window before four years of data exist), shown with the rolling correlation in Figure 8. Before 2020 the correlation was 0.0–0.3 and the beta was unstable, near 3 in 2013–2015 only because Bitcoin was so volatile and about zero over the four years to early 2020; since 2022 the correlation has been 0.3–0.6 and the beta 1.6–2.2.

The rise in co-movement coincides with institutional entry, and is documented independently by Copestake, Furceri, and Terracciano (2026). The forecast of Section 5 uses a beta of 1.5–2.0.

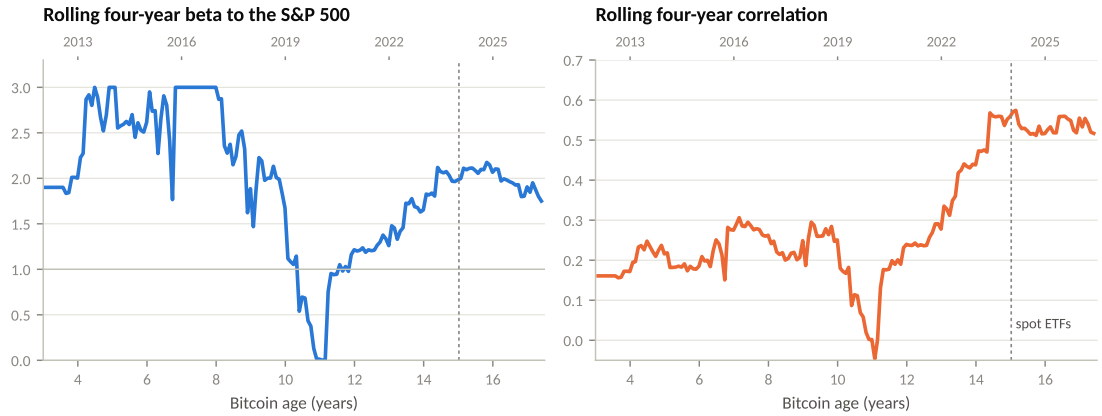


Figure 8: Bitcoin's rolling four-year beta and correlation with the S&P 500 (month-end log returns; expanding window before four years of data). The dotted line marks spot-ETF approval.

A.2 Difference from the companion paper. The companion paper reports betas of 0.33 (four years), 1.06 (since 2016) and 2.12 (since 2012), with correlations of 0.07–0.18. We cannot reproduce the two recent figures on month-end data. We can reproduce figures close to them (0.57, 1.09, 1.62; correlations 0.13–0.21) by pairing month-end Bitcoin prices with monthly-average index levels, a half-month misalignment that attenuates the covariance, and we conclude that the earlier estimates were attenuated in that way. The estimates here supersede them; the companion paper's CAPM range of 6–18% becomes 11–16%.

A.3 The premium, and two approximations. The equity premium is 4%, with 6% as a high case, as in the companion paper; each point of premium moves Bitcoin's required return by about 1.75 points. Two approximations in the market term are worth stating. First, for a holder with weight w in Bitcoin and the rest in the market, the exact term is $(1-w)\beta$ times her price of market variance; at w of 8–19% the factor $(1-w)$ lowers the term by about a point, which the stated beta range absorbs. Second, a holder with risk aversion γ prices market variance at $\gamma\sigma_m^2$, which at $\gamma = 1.7$–2.8 and $\sigma_m = 16\%$ is 4.4–7.2%, bracketing the 4% and 6% premia used; the market term and the concentration term are therefore priced consistently.

B. CONCENTRATION

The holder weight w is the input the paper measures least directly, so this appendix shows how it is built (Figure 9). One number is measured: in 2021Q3 US households that owned any crypto held 19% of their financial assets in it on average (Candia, Coibion, Gorodnichenko, and Weber 2026), with the Austrian 2023 survey (OeNB 2025) as a cross-check. Everything else follows from market clearing, capitalization $= \bar{w}\times$ participating wealth, run backward and forward from that anchor.

B.1 Participation. The participant count is the share of US adults holding crypto in household surveys: 0.4% (2014) and 1.2% (2018) from the Federal Reserve's payment-choice surveys, 11% in 2021Q3 from Candia et al., 12–13% since from the Fed's SHED and Pew. The path is interpolated log-linearly between survey dates; the Coin Metrics count of addresses holding a balance supplies the shape only before the first survey point. Panel (a) also shows the alternative of scaling the address count to the 2021 point, which

overstates participation five-fold in 2018 (6.4% against 1.2%) because addresses multiply faster than people; it is not used.

B.2 Wealth behind the holders. Participating wealth is anchored at capitalization divided by 0.19 in 2021Q3 (about $4.3 trillion) and moved with the participant count times wealth per participant, a ramp from a quarter of the 2021 level in 2012 to 1.4 times by 2026 (Appendix D.5 replaces the ramp with US net worth per adult). From 2024 a second class of holders enters through the spot ETFs and public-company treasuries, about 2.5 million BTC by mid-2026 (roughly 1.25 million in US spot ETFs and 1.26 million in corporate treasuries, Strategy alone about 0.85 million), worth about $190 billion at the May 2026 price. Their investors are not "crypto owners" in household surveys and hold small weights; at 1.5% of wealth they represent about $13 trillion of participating wealth, at 3% about $6 trillion, against $7 trillion for survey-type holders (panel b).

B.3 The weight. Dividing capitalization by participating wealth gives the implied average weight (panel c): about 3% in 2013, 17% in 2017, 18% in 2021, and in 2026 either 21% for survey-type holders alone or 8% once the institutional channel is counted at 1.5%. The path spikes at the 2013, 2017 and 2021 peaks because capitalization moves with the price while participation is smooth; that is the feature Appendix D.4 documents as a negative predictor and the reason the paper treats the weight as an input to a date-by-date restriction rather than a signal. The band used in Section 6 runs from the diluted back-cast (lower edge) to the all-in holder of the option bound (0.9, upper edge); the 2026 range of Section 5, $w \in [0.08, 0.19]$, is the diluted value up to the surveyed value. Two omissions work in opposite directions and are left as they are: coins that are permanently lost (estimates of 3–4 million BTC) should be removed from capitalization, which would lower every weight by 15–20%, and the wealth of the roughly 15% of owners who hold more than half their portfolio in crypto is understated by an average, which would raise the weight of the marginal holder if she is drawn from that group.

B.4 Risk aversion and the option bound. The other two ingredients of the concentration term are taken from outside the paper. Risk aversion is the household-finance range of Section 2, $\gamma \in [1.7, 2.8]$; Section 5 explains why it is not calibrated to holders' stated expectations. The upper edge of the historical band is Martin's (2017) bound: for a holder with all her wealth in the asset and log utility ($\gamma w = 1$), the required excess return is the risk-neutral variance, 66–84 points a year on Deribit options over 2017–2022 (Almeida, Grith, Miftachov, and Wang 2024) and 16–30 points at the implied volatilities of 40–55% observed since. It is a requirement for an all-in holder, not a forecast for the asset; the band's upper edge scales it down after 2020 as the holder base dilutes.

C. MORTALITY

C.1 The curated elite death table. All 36 deaths among the 219 ever-top-20 assets, with the classification each received. *Type*: jump = sudden, value-destroying collapse priced as a Poisson event; exchange = failure of the issuing exchange; fade = gradual abandonment and loss of listings; contested = flagged by a mechanical rule but the chain still operates. δ is the approximate fraction of pre-event value recoverable by a holder (last liquid price or successor claim as a share of pre-event value); it is an estimate, not an accounting figure. *Conf.* flags classification confidence (H/M/L). Dates are the curated economic death dates; mechanical rules alone would misdate LUNA, UST, and FTT, whose collapses are entered from the public record.

Asset	Death date	Type	δ	Note	Conf.
LTL	2014-12	fade	$\sim$0	minor 2013-era coin	M
FCT (Factom)	2021-07	fade	$\sim$0.05	sponsor company bankrupt 2020; chain stalled	H
LBC (LBRY)	2021-09	fade/regulatory	$\sim$0.05	SEC action 2021; project wound down	H
PPT (Populous)	2021-11	fade	$\sim$0	abandoned	M
ETP (Metaverse)	2022-01	fade	$\sim$0	abandoned	M
NCASH (Nucleus Vision)	2022-01	fade	$\sim$0	delisted majors 2021	H
ZB	2022-02	exchange	$\sim$0	ZB.com halted withdrawals 2022	H
WTC (Waltonchain)	2022-04	fade	$\sim$0	abandoned	H
UST (TerraUSD)	**2022-05-12**	**jump**	$\sim$0.02	algorithmic stablecoin depeg	H
LUNA (Terra)	**2022-05-12**	**jump**	$\sim$0.01	supply hyperinflation; LUNA2 airdrop gave partial recovery	H
QASH (Liquid)	2022-07	exchange	$\sim$0	exchange hacked 2021, absorbed into FTX	H
QC (Qcash)	2022-08	fade	$\sim$0	abandoned	M
FTT (FTX Token)	**2022-11-10**	**jump**	$\sim$0.05	FTX insolvency	H
MITH (Mithril)	2023-03	fade	$\sim$0	serial delistings	H
QSP (Quantstamp)	2023-05	fade	$\sim$0.02	token abandoned	M
SEELE	2023-07	fade	$\sim$0	abandoned	M
VTC (Vertcoin)	2023-08	fade	$\sim$0.02	lost major listings	M
MDA (Moeda)	2023-09	fade	$\sim$0	abandoned	M
BTS (BitShares)	2024-06	fade	$\sim$0.02	community remnant only	M
AE (Aeternity)	2024-10	fade	$\sim$0.02	team wind-down	M
BCN (Bytecoin)	2024-10	fade	$\sim$0	long dormant; delisted majors years earlier	H
OMG (OMG Network)	2025-02	fade	$\sim$0.02	Binance delisting; development ceased	H
KEY (SelfKey)	2025-04	fade	$\sim$0	abandoned	M
LOOM (Loom Network)	2025-05	fade	$\sim$0	abandoned	H
XEM (NEM)	2025-05	fade	$\sim$0.05	major delistings; earlier opt-in XYM airdrop (2021)	H
BTG (Bitcoin Gold)	2025-06	fade	$\sim$0.02	Binance delisting	H
LINA (Linear)	2025-06	fade	$\sim$0	project shut down	M
PDA (PlayDapp)	2025-06	fade	$\sim$0.02	post-2024-exploit decline (PLA$\rightarrow$PDA swap preceded)	M
YFII (DFI.Money)	2025-07	fade	$\sim$0	abandoned fork	M
FLOW	2025-09	**contested**	n/a	chain operating; >99% below ATH	H
MXC	2025-09	fade	$\sim$0.02	rebranding/decline	L
STMX (StormX)	2025-10	fade	$\sim$0	abandoned	M
NANO	2025-12	**contested**	n/a	active community; lost major listings	M
ICP (Internet Computer)	2026-01	**contested**	n/a	alive, top-100; "death" is an artifact of the listing-day price-spike ATH	H
HIFI	2026-01	fade	$\sim$0	delisted	L
SUSHI (SushiSwap)	2026-03	**contested**	n/a	protocol operating; >99% below ATH	H

Headline sensitivities. All 36 deaths: hazard 3.54%/yr (CI 2.48–4.90). Excluding the four contested cases: 3.15%/yr (CI 2.15–4.44). Jump-collapses only (LUNA, UST, FTT): 0.30%/yr (CI 0.06–0.86). In the ever-top-50 universe the jump set adds SRM (Serum) and ANC (Anchor), both 2022 casualties of the same two cascades, for 0.23%/yr.

Migrations excluded as deaths (holders received successor value): MATIC→POL, EOS→Vaulta (A), FTM→S (Sonic), GNT→GLM, RNDR→RENDER, BTT redenomination, NU→T (Threshold merger), STRAT→STRAX, MAID→eMAID, HT→HTX, REP→REPv2, AGIX→FET (ASI merger), XZC→FIRO, TOMO→VIC, NPXS→PUNDIX, BNX redenomination, and DGD, whose dissolution refunded the ETH treasury to holders: a death with δ near 1, the mirror image of LUNA.

C.2 The Lindy test. Poisson regression of monthly death indicators on log age, offset $\log(1/12)$, standard errors clustered by asset. Lindy (hazard $= k/\text{age}$) requires a log-age coefficient of -1.

Death definition	Deaths	γ_{age}, no FE	γ_{age}, year FE
Strict (feed termination)	234	+0.56** (0.07)	+0.44** (0.06)
Broad (feed + price)	263	+0.69** (0.07)	+0.56** (0.07)
Abandonment (all mechanical)	588	–	+0.51** (0.04)
Curated (final)	**520**	–	+0.42** **(0.04)**
Curated, 2025 feed-wall removed	–	–	+0.65** (0.08)
Curated, birth-cohort FE in place of year FE	520	–	+1.62** (0.10)
Elite (ever top-20) subsample	36	–	+0.68** (0.21)

Standard errors in parentheses, clustered by asset. ** significant at the 1% level; * at the 5% level. Every coefficient differs from the Lindy value of -1 at the 1% level.

The log-survival–log-age slope over ages 1–10 runs from -0.21 (strict) to -0.49 (curated), i.e. a cumulative Pareto constant $\hat{k} \approx 0.2$–0.5 across definitions, an order of magnitude below the 5.6 the age rule would require. Two further robustness checks address the standard identification worry that an age effect might be a disguised cohort effect (age, cohort, and period are collinear, so only two can enter at once): replacing calendar-period fixed effects with birth-cohort fixed effects *raises* the age elasticity to $+1.62$ (s.e. 0.10), so the rejection of Lindy is not an artifact of the period controls, and the period-FE $+0.42$ is the conservative figure. Parametrically, a Weibull fit to the curated durations gives shape $\rho = 1.57$ (95% CI 1.47–1.67), where $\rho > 1$ means a hazard rising with age and Lindy requires $\rho < 1$; a log-logistic fit (slightly preferred by AIC, 3,342 vs 3,371) implies a hump-shaped hazard that rises through the first decade of life. The implied median lifetime of a core-universe cryptoasset is 6.4 years under both the Weibull and the Kaplan–Meier estimates.

C.3 Death-definition sensitivities. Volume-collapse deaths require an absolute floor ($250k/day trailing volume) in addition to the 1%-of-peak rule; without the floor, 2021-mania base effects falsely kill operating mid-caps. The elite hazard is 3.54% with all curated deaths, 3.15% excluding the four contested zombies, and 2.36% (CI 1.51–3.51%) treating all twelve 2025 deaths as censoring; the Lindy rejection holds under every variant (C.2).

C.4 Forward-looking hazards the graveyard cannot see: the quantum threat. The hazard measured here is a backward-looking frequency, and some risks have never yet fired for anyone. The clearest is quantum computing: a cryptographically relevant quantum computer running Shor's algorithm would break the elliptic-curve signatures that secure Bitcoin, and the event would be exactly what the marginal-holder condition calls a jump, a λ-event with its own loss fraction. The model therefore *covers* the quantum threat by construction; what the graveyard cannot do is price it, because the realized death count contains no quantum events. Two observations discipline it. First, the loss given the event is scenario-dependent, not automatically total: roughly a quarter of Bitcoin's supply sits in addresses with exposed public keys, including the presumed-Satoshi coins (Deloitte's standing estimate is about 4 million BTC), and is the immediately vulnerable part, while a telegraphed advance in quantum hardware would allow a migration to post-quantum

signatures that preserves most other value; so δ ranges from near zero (a surprise break) to substantial (an orderly transition). Second, if the market prices a material quantum hazard, it must appear where all jump risk appears, in deep out-of-the-money put prices. The option-implied crash intensity is therefore the running referee between the graveyard's 0.3% and any larger forward-looking hazard, quantum included; at this writing it shows no such gap. An investor who privately assigns, say, a 1% annual probability to a disorderly quantum break with half of value lost should simply add 0.5 points to the mortality term; the equation accepts the view without modification.

D. ASSEMBLY AND VALIDATION

D.1 Propagating input uncertainty. The 14–25% envelope of Section 5 sums the extremes of four input ranges: beta 1.5–2.0 on a 4% equity premium; risk aversion $\gamma \in [1.7, 2.8]$; holder weight $w \in [0.08, 0.19]$; mortality term 0.3–3 points; with the risk-free rate at 5.0% and $\sigma = 0.42$. To obtain a distribution rather than an envelope we draw each input independently and uniformly over its range, 400,000 draws, and compute the required return $r_f + \beta\,\mathrm{ERP} + \gamma\sigma^2 w + \lambda(1-\delta)$ for each. The median is 18.9%, the central 68% interval 17.2–20.9%, and the 90% interval 16.2–22.1%. The uniform assumption is deliberately uninformative; any distribution concentrated toward the interior of the ranges tightens the result. Narrowing the mortality term to 0.3–1.5 points moves the median to 18.2% and the 90% interval to 15.7–21.1%; narrowing the holder weight to 0.12–0.19 gives 19.8% and 17.3–22.4%. The same procedure gives Table 5 and the gold figures of Appendix E (beta uniform on $[-0.1, 0.1]$, σ on $[0.13, 0.20]$, w on $[0.02, 0.10]$, mortality on 0–0.3 points: median 5.5%, 90% interval 5.0–6.0%) and Bitcoin's digital-gold endpoint ($\sigma = 0.15$, w on $[0.02, 0.10]$, beta and mortality at Bitcoin's ranges: median 14.0%, 90% interval 12.3–15.6%). The script is `23_uncertainty.py` in the replication package.

D.2 Units. Equation (1) is a condition on the arithmetic drift μ. The compound growth rate of a lognormal price is $\mu - \sigma^2/2$, so the same asset has a drift and a CAGR that differ by half its variance: 9 points at 42% volatility, 32 at 80%, over 100 at 150%. Section 6 therefore compares the required return with the realized drift, estimated over each window as twelve times the mean monthly log return plus half the annualized variance of monthly log returns. Comparing with the CAGR instead would understate delivery by the drag; over the 2021Q3–May 2026 window the CAGR is 13% and the drift 27%.

D.3 The backtest in matching units. Over the 113 monthly five-year windows starting January 2012 to May 2021, realized drift minus the required return averages $+60$ points at the band's lower edge, -49 at its upper edge, and $+5$ at its midpoint (median $+21$); realized drift lies inside the band in 48% of windows. As a forecast of each window the midpoint has a mean absolute error of 51 points and a correlation with realized drift of 0.14; the age rule has 21 points and 0.83, and is closer in 91% of windows. The required return passes a test of level and fails a test of timing, for the reason in D.4. A rate can be the correct required return and a poor forecast of when returns arrive; the companion paper's backtest, which selects on fit to the realized path, cannot distinguish an unbiased cost of capital from a smooth curve fitted to that path.

D.4 The concentration quantity is not a timing signal. Forward h-year log growth is regressed on the real-time concentration quantity $x = \sigma^2 w$ (with w the back-cast weight) and the age-rule regressor $z = \ln(1 + h/A)/h$; heteroskedasticity- and autocorrelation-robust standard errors with $12h+6$ lags; overlapping monthly windows from 2012 (independent observations $\approx$ 2–4, so the t-statistics are descriptive).

The concentration quantity enters with a negative sign: it is highest at market peaks, because capital-

Horizon	x alone	z alone	Joint: x	Joint: z	R^2 (joint)
3y	-0.8 (t -4.7)	$+4.9$ (t 5.0)	-0.8	$+5.1$	0.57
5y	-0.8 (t -5.9)	$+7.0$ (t 13.5)	-0.8	$+6.8$	0.82

ization sits in the numerator of w and trailing variance is highest after a run-up, and returns after peaks are low. The age regressor reproduces the companion paper's coefficient (5–7, the power-law exponent). This is why the paper treats the marginal-holder condition as a date-by-date restriction and never as a forecasting variable; it is also why the age rule's in-sample fit should not be read as evidence of a priced mechanism.

D.5 Wealth-path robustness. The back-cast's wealth-per-participant ramp (a quarter of the 2021 level in 2012, 1.4 times by 2026) is a judgment about who held Bitcoin. Replacing it with the growth of US household net worth per adult (Federal Reserve Z.1, about $80T in 2013 to $170T in 2026), so that early holders are assumed as wealthy as the average American, lowers the early back-cast weight and the band's lower edge: 2013 becomes 21–231% (baseline 26–231%), 2017 (28–67%) and 2021Q3 (23–47%, the anchor) are unchanged, and 2026 becomes 14–29% (baseline 14–27%). The average gap between realized drift and the band midpoint moves from $+5$ to $+7$ points.

E. EXTENSIONS

E.1 Gold, as a check on the equation. Gold is the natural comparison, both because Bitcoin is so often called digital gold and because gold is the one other large asset that pays no income, so the whole of its return must come from price. Equation (1) prices it with the same four inputs, every one of which sits at the opposite end of the range from Bitcoin's. Market risk: gold's correlation with equities is close to zero over long samples (Erb and Harvey 2013), so the beta term is a few tenths of a point either way ($\beta \in [-0.1, 0.1]$). Concentration: gold is the most widely held asset on earth, about 224,000 tonnes above ground, of which 44% is jewelry, 18% central-bank reserves, and 23% private bars, coins, and ETFs (World Gold Council 2026), worth about $29 trillion at today's price, some 6% of global household wealth; its marginal holder is a diversified investor with a few percent in it, not an enthusiast with a fifth. With volatility of 13–20% (13.1% over 1836–2011 in Barro and Misra 2016; higher in the last two years), a holder weight of 2–10%, and $\gamma \in [1.7, 2.8]$, the concentration term is 0.1–1.1 points, median 0.3. Mortality: gold has no failure mode that destroys the asset; the closest analogues are sovereign restrictions, such as the United States' compulsory purchase of private gold in 1933 at the old official price before the dollar was devalued, which are jurisdiction risks to a holder rather than deaths of the asset, and we allow 0–0.3 points. The condition returns a required return for gold of about 5.5% a year (90% interval 5.0–6.0%; envelope 4.7–6.8%; Appendix D.1): the risk-free rate plus half a point, a Sharpe ratio near zero. This is a reproduced result, and that is its value here: Barro and Misra (2016) find that gold's real price appreciation averaged 1.1% a year over 175 years, with a covariance with consumption indistinguishable from zero, and conclude that its expected return should sit close to the risk-free rate; the equation reaches the same place from observable portfolios rather than from a consumption model. Gold's recent history is instructive. From the end of 2023 to November 2026 gold compounded at about 27% a year (from about $2,060 to about $4,040 an ounce, with a gain of some 65% in 2025 alone and a flat twelve months since; LBMA 2026), roughly twenty points a year above anything its holders could have required. That excess is news, not a payment for risk: central-bank purchases, fiscal and

geopolitical developments, and ETF inflows the market had not priced. It is also a reminder that three years of realized returns say little about a required return, for gold as for Bitcoin. In equilibrium the equation puts gold's expected return from here back at 5–6% nominal; extrapolating 2024–25 would be the gold version of the age rule.

E.2 What "digital gold" would mean. The comparison also fixes the phrase in return terms. If Bitcoin's volatility fell to gold's 15% and its holder base diluted to gold's few percent per holder, the concentration term would all but vanish, as gold's has, and what remained would be the two terms gold does not carry: a beta of 1.5–2.0 and a mortality term of 0.3–3 points. The condition then gives about 14% a year (90% interval 12–16%; Appendix D.1). Only if those two also converged to gold's, with Bitcoin as uncorrelated with equities as gold and as immune to failure, would its required return reach gold's 5–6%. On the evidence of Appendix A, Bitcoin has been moving away from gold in that respect, not toward it: integration has lowered its concentration premium and raised its market risk. Digital gold names a required return near the risk-free rate; Bitcoin's is converging instead on that of a high-beta equity.

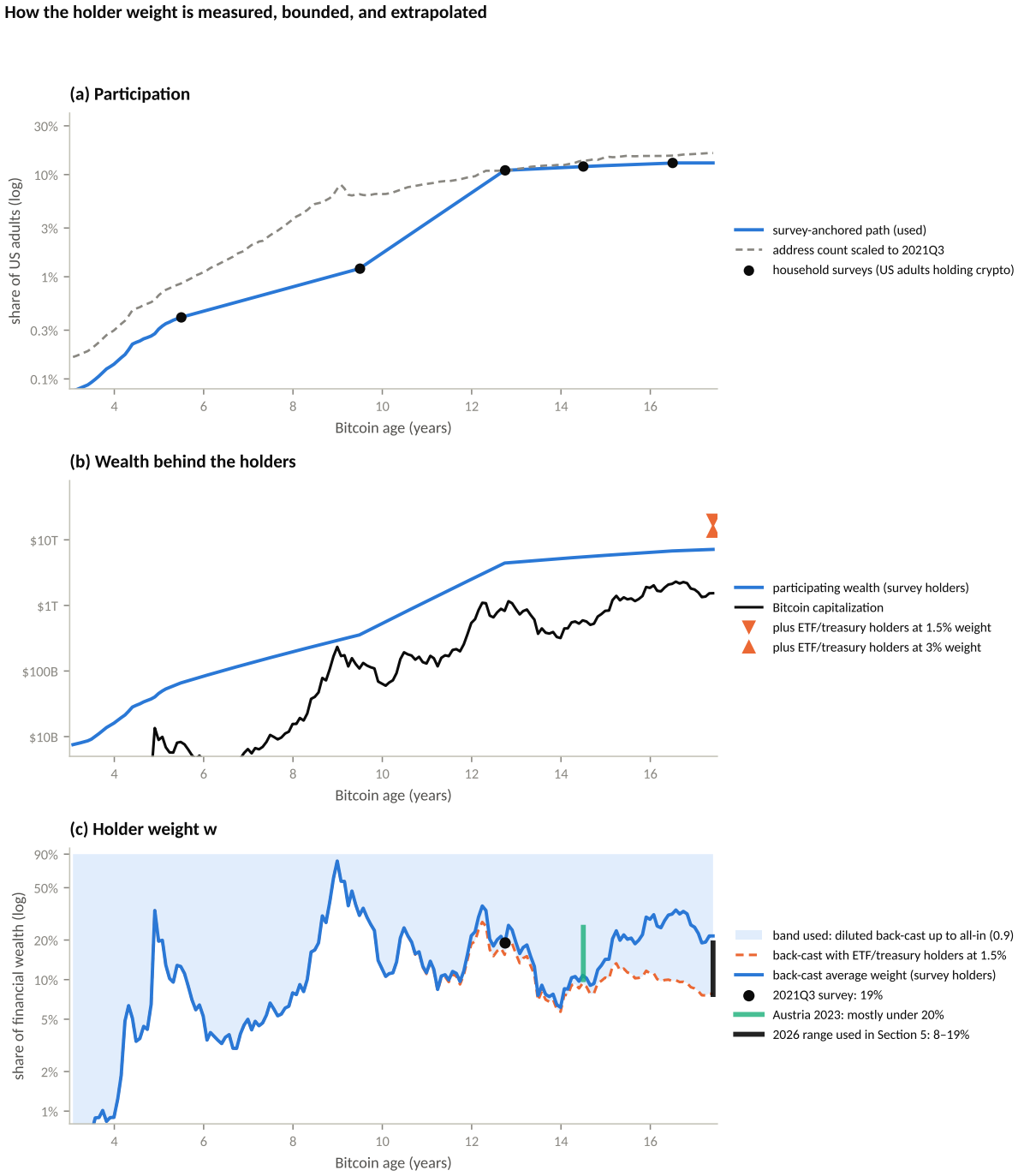


Figure 9: Constructing the holder weight. (a) Share of US adults holding crypto: survey points, the log-linear path used, and the address-count alternative not used. (b) Participating wealth of survey-type holders, Bitcoin's capitalization, and the 2026 additions for ETF and treasury investors at 1.5% and 3% weights. (c) The implied average weight, the 2021Q3 survey value, the Austrian cross-check, the diluted back-cast, and the band used in Section 6; all panels on logarithmic scales.